AI-POWERED SMART INSURANCE CLAIM MANAGEMENT SYSTEM

A MINI PROJECT REPORT

Submitted by

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Under the Guidance of

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DEPARTMENT OF COMPUTER APPLICATIONS

MANGALAM COLLEGE OF ENGINEERING, ETTUMANOOR

(Affiliated to APJ Abdul Kalam Technological University)

OCTOBER 2026

VISION

To become a centre of excellence in computer applications,competent in the global ecosystem with technical knowledge,innovation with a sense of social commitment.

MISSION

- To serve with state of the art education,foster advanced research and cultivate innovation in the field of computer applications.
- To prepare learners with knowledge skills and critical thinking to excel in the technological landscape and contribute positively to society.

Program Educational Objectives

- PEO I :Graduates will possess a solid foundation and in-depth understanding of computer applications and will be equipped to analyze real-world problems, design and create innovative solutions, and effectively manage and maintain these solutions in their professional careers.
- PEO II: Graduates will acquire technological advancements through continued education, lifelong learning and research, thereby making meaningful contributions to the field of computing.
- PEO III: Graduates will cultivate team spirit, leadership, communication skills, ethics, and social values, enabling them to apply their understanding of the societal impacts of computer applications effectively.

Program Specific Outcomes

- **PSO I**: Apply advanced technologies through innovations to enhance the efficiency of design and development.
- **PSO II**: Apply the principles of computing to analyze, design and implement sustainable solutions for real world challenges.

MAPPING OF PO-PSO-SDG

1. MAPPING WITH PROGRAM OUTCOMES (POs):

SL.NO	POs ADDRESSED	RELEVANCE TO PROJECT
1	PO1	The system applies programming, database and web technologies to develop an online insurance claim management system.
2	PO2	The project analyzes the delays, paperwork and fraud risks involved in manual insurance claim processing.
3	PO3	A web-based system is designed and developed for claim submission, document upload, fraud screening, claim verification and status tracking.
4	PO4	Modern tools such as Django, Scikit-learn, EasyOCR and AI-based image analysis are used to build and test the system.
5	PO5	The project involves planning, development and testing activities that require effective individual work and coordination of different system components.
6	PO6	Project management principles such as planning, scheduling, task allocation, resource management and systematic development are applied throughout the project.
7	PO7	Role-based access, administrator approval of users and companies, and controlled access to claim documents support ethical handling of sensitive data.
8	PO8	Developing the system required learning contemporary technologies such as machine learning, OCR and AI-based vehicle damage analysis.

LIST OF PROGRAM OUTCOMES (POs):

PO1 (**Foundation Knowledge**): Apply knowledge of programming logic, database concepts and coding fundamentals for solution architecture and problem solving.

PO2 (Problem Analysis): Identify, review, formulate and analyze problems in traditional insurance claim processing with focus on user requirements.

PO3 (Development of Solutions): Design and develop an innovative online insurance claim management solution for efficient complaint management and tracking.

PO4 (Modern Tool Usage): Select and apply modern computational tools and technologies such as Python, Django, MySQL, JavaScript and machine learning for system development.

PO5 (**Individual and Teamwork**): Function and communicate effectively as an individual or team member during the planning, development and testing of the project.

PO6 (**Project Management and Finance**): Apply project management principles such as planning, scheduling, resource management and work breakdown structure.

PO7 (Ethics): Follow professional ethics and cybersecurity practices to protect user information, claim data, uploaded documents and digital evidence from unauthorized access.

PO8 (Life-long learning): Develop change management skills and continuously learn and adapt to contemporary technologies and methods for improving the insurance claim management system.

2. MAPPING WITH PROGRAM SPECIFIC OUTCOMES (PSOs):

LIST OF PROGRAM SPECIFIC OUTCOMES (PSOs):

SL.NO	PSOs ADDRESSED	RELEVANCE TO PROJECT
1	PSO1 – Apply advanced technologies through innovations to enhance the efficiency of design development	Advanced technologies such as machine learning, OCR and AI-based vehicle damage analysis were applied to design an efficient claim management and fraud detection platform.
2	PSO2 – Apply the principles of computing to analyze, design and implement sustainable solutions for real-world challenges	Computing principles of data modelling, database design, security and software design were applied to build a sustainable digital solution for insurance claim processing.

PSO 1: Apply advanced technologies through innovations to enhance the efficiency of design development.

PSO 2: Apply the principles of computing to analyze, design and implement sustainable solutions for real world challenges.

3. MAPPING WITH SUSTAINABLE DEVELOPMENT GOALS (SDGs):

SDG NO	SDGs ADDRESSED	RELEVANCE TO PROJECT
SDG 8	Decent Work and Economic Growth	The platform automates claim handling and reduces manual effort, supporting efficient and productive insurance services.
SDG 9	Industry, Innovation and Infrastructure	The project applies machine learning, OCR and AI image analysis to create an innovative digital infrastructure for insurance claim management.
SDG 16	Peace, Justice and Strong Institutions	The fraud detection and verification workflow promotes transparent, accountable claim handling and helps reduce insurance fraud.

SUSTAINABLE DEVELOPMENT GOALS (SDGs):

SDG 1 – No Poverty: End poverty in all its forms everywhere.

SDG 2 – Zero Hunger: End hunger, achieve food security and improved nutrition, and promote sustainable agriculture.

SDG 3 – Good Health and Well-Being: Ensure healthy lives and promote well-being for all at all ages.

SDG 4 – Quality Education: Ensure inclusive and equitable quality education and promote lifelong learning opportunities for all.

SDG 5 – Gender Equality: Achieve gender equality and empower all women and girls.

SDG 6 – Clean Water and Sanitation: Ensure availability and sustainable management of water and sanitation for all.

SDG 7 – Affordable and Clean Energy: Ensure access to affordable, reliable, sustainable and modern energy for all.

SDG 8 – Decent Work and Economic Growth: Promote sustained, inclusive and sustainable economic growth, full and productive employment and decent work for all.

SDG 9 – Industry, Innovation and Infrastructure: Build resilient infrastructure, promote inclusive and sustainable industrialisation and foster innovation.

SDG 10 – Reduced Inequalities: Reduce inequality within and among countries.

SDG 11 – Sustainable Cities and Communities: Make cities and human settlements inclusive, safe, resilient and sustainable.

SDG 12 – Responsible Consumption and Production: Ensure sustainable consumption and production patterns.

SDG 13 – Climate Action: Take urgent action to combat climate change and its impacts.

SDG 14 – Life Below Water: Conserve and sustainably use the oceans, seas and marine resources.

SDG 15 – Life on Land: Protect, restore and promote sustainable use of terrestrial ecosystems, manage forests sustainably, combat desertification and halt biodiversity loss.

SDG 16 – Peace, Justice and Strong Institutions: Promote peaceful and inclusive societies, provide access to justice for all and build effective, accountable and inclusive institutions.

SDG 17 – Partnerships for the Goals: Strengthen the means of implementation and revitalise the global partnership for sustainable development.

MANGALAM COLLEGE OF ENGINEERING, ETTUMANOOR
DEPARTMENT OF COMPUTER APPLICATIONS
OCTOBER 2026



DECLARATION

I hereby certify that the work presented in the project entitled ***"AI-Powered Smart Insurance Claim Management System"*** *submitted to the* ***DEPARTMENT OF COMPUTER APPLICATIONS*** *is an authentic record of my own work carried out under the supervision of* ***Ms. Divya S B, HOD & Assistant professor****. This work has not been submitted to any other institution or university for the award of any other degree. The report has been checked for plagiarism by the college and the similarity index is within the permissible limit set by the college.*

Name & Signature of Student

ARCHA DIJIKUMAR

Date:

Place:

MANGALAM COLLEGE OF ENGINEERING, ETTUMANOOR
DEPARTMENT OF COMPUTER APPLICATIONS
OCTOBER 2026



CERTIFICATE

Certified that the report entitled ***"AI-Powered Smart Insurance Claim Management System"*** *submitted by* ***ARCHA DIJIKUMAR (MLM25MCA-2021)*** *to the* ***APJ Abdul Kalam Technological University*** *in partial fulfilment of the requirements for the award of* ***the Degree of Master of Computer Applications*** *is a bonafide record of the* ***MINI PROJECT*** *work carried out by the student under our guidance and supervision during the academic year 2026-27.*

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ABSTRACT

Insurance claim processing in most organizations is still slow, paper-based and heavily dependent on manual verification. Claim officers must check documents by hand, data entry mistakes are common, and customers have little visibility into the status of their claims. Fraudulent claims add to these problems. Rule-based checks cannot catch advanced fraud patterns, and the resulting financial losses raise costs for insurers and honest policyholders alike.

The AI-Powered Smart Insurance Claim Management System is a web-based platform that addresses these problems by automating the claim process end to end. It uses Artificial Intelligence, Machine Learning, Optical Character Recognition (OCR) and image processing to make claim handling faster, more accurate and more transparent. The system follows a role-based approach with three types of users: policyholders, insurance companies and administrators.

Users can register, submit claims through online forms, upload supporting documents and vehicle images, and track claim status in real time. When a document is uploaded, OCR extracts the text and key fields automatically, which removes manual data entry. A machine learning model then analyzes the claim details and predicts whether the claim is fraudulent, along with a fraud probability score that serves as a risk indicator. For vehicle claims, users upload photographs of the vehicle from six sides, and the system analyzes them to produce a damage description that supports the assessment.

Insurance companies get a single dashboard to review claims together with the fraud prediction, extracted document data and damage analysis. They can approve or reject claims, verify or reject uploaded documents with remarks, and manage their policies. Administrators approve user registrations, review registered companies and users, and respond to complaints.

By combining document intelligence, fraud prediction and image-based damage analysis in one platform, the system reduces manual effort, minimizes human error, speeds up decisions and improves the accuracy of fraud detection. Overall, it improves efficiency, transparency and customer satisfaction in insurance claim management.

Keywords: Insurance Claim Management, Fraud Detection, Machine Learning, OCR, Image Processing, Damage Analysis, Python, Django, Scikit-learn, EasyOCR, MySQL, HTML, CSS, Bootstrap, JavaScript

Mapping with Sustainable Development Goals (Mention the Goal)	SDG 8 Decent Work and Economic Growth SDG 9 Industry, Innovation and Infrastructure SDG 16 Peace, Justice and Strong Institutions

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LIST OF ABBREVIATIONS

ABBREVIATION		FULL FORM
AI	—	Artificial Intelligence
API	—	Application Programming Interface
CSS	—	Cascading Style Sheets
DB	—	Database
DBMS	—	Database Management System
DFD	—	Data Flow Diagram
DOCX	—	Microsoft Word Open XML Document
HTML	—	HyperText Markup Language
HTTP	—	HyperText Transfer Protocol
IDE	—	Integrated Development Environment
JPG	—	Joint Photographic Experts Group image format
ML	—	Machine Learning
MVC	—	Model-View-Controller
MySQL	—	My Structured Query Language
OCR	—	Optical Character Recognition
PDF	—	Portable Document Format
PNG	—	Portable Network Graphics
PO	—	Program Outcome
PSO	—	Program Specific Outcome
SDG	—	Sustainable Development Goal
SQL	—	Structured Query Language
SVM	—	Support Vector Machine
UI	—	User Interface
URL	—	Uniform Resource Locator

CHAPTER 1
INTRODUCTION

1.1 Background

Insurance claim processing is an important part of the insurance sector, particularly for vehicle and similar policies where a claim must be supported by documents, evidence of damage, and verification by the insurer. In practical situations, a policyholder has to submit claim details, supporting documents, and photographs, and then wait for the insurance company to review them manually before a decision is made.

Traditional claim management often depends on paper forms, scanned documents sent by email, telephone follow-ups, and manual inspection of damage and records. These methods are slow and do not provide a unified record of claims, claim status, damage evidence, fraud checks, and communication between the customer and the company. Manual verification also makes it difficult to detect fraudulent or manipulated claims consistently.

The AI-Powered Smart Insurance Claim Management System is proposed as a centralized web-based platform for submitting, reviewing, and settling insurance claims with the support of artificial intelligence. It provides a structured environment where users can register, submit claims, track claim status, and communicate with insurance companies, while companies can review claims and manage their decisions.

The system also incorporates AI-based features such as image-based vehicle damage analysis, OCR-assisted extraction of data from uploaded claim documents, and machine-learning-based fraud prediction. These features are intended to reduce manual effort, speed up claim review, and help insurers identify suspicious claims early.

1.2 Motivation

The motivation for this project comes from the delay, effort, and uncertainty involved in conventional claim handling. A claim portal alone may record a request, but a useful claim management system should also help the insurer assess the damage, read the submitted documents, estimate the likelihood of fraud, and keep a verifiable record of every decision.

Another motivation is the growing problem of insurance fraud, which increases costs for insurers and honest customers. Manual checks are time-consuming and depend on individual judgement, so there is a need for an assisted approach that gives reviewers a prediction and a probability while leaving the final decision to a human. The system therefore treats AI output as decision support. The company

verifies or rejects each document with a recorded note, and access is divided by role so that users, companies, and administrators see only the functions meant for them.

1.3 Scope & Objectives

The scope of the system covers a web-based platform for claim submission, review, and fraud detection. It includes user and insurance company registration and login, administrator approval of users, policy management by companies, claim submission and status tracking, vehicle damage image upload and AI-based analysis, fraud document upload with OCR-based extraction, fraud prediction, complaint handling, and administrative management.

The prototype uses a Django-based web architecture with HTML, CSS, JavaScript, and Bootstrap on the frontend and a MySQL database on the backend. Scikit-learn is used for fraud prediction, and EasyOCR is used for text extraction from documents. The system has three roles: user, insurance company, and administrator.

The main objective is to develop a centralized platform that allows users to submit claims and track their status, and allows insurance companies to review, accept, or reject claims. The system enables users to upload six vehicle images (front, back, left, right, top, and bottom) so that the company can run an AI-based damage analysis. It supports OCR-based extraction of text and key fields from uploaded PDF, DOCX, and image documents, and a machine-learning model that predicts whether a claim is fraudulent along with a probability percentage. Insurance companies can verify or reject fraud-checked documents with a note, and download documents and results. A complaint and reply mechanism connects users and companies with the administrator, and the administrator can approve or reject users and monitor the system through a dashboard. The system is designed modularly so that further notification services, payment integration, and additional analysis features can be added in future.

1.4 Problem Statement

Policyholders who raise insurance claims face delays, repeated document submissions, and a lack of clear information about the status of their claim. Claim-related information is often scattered across paper forms, emails, photographs, and phone conversations, making it difficult to track claims efficiently and maintain an organized history of decisions.

Insurance companies, on the other hand, must manually inspect damage photographs, read supporting documents, and judge whether a claim is genuine. This is slow and error-prone, and fraudulent claims can pass unnoticed because the checks are inconsistent and not supported by data.

There is also a need for controlled and trustworthy handling of sensitive information. Claim documents contain personal, financial, and vehicle details, so access must be restricted by role, and users and companies must be verified before they use the system.

The AI-Powered Smart Insurance Claim Management System addresses these challenges by integrating claim submission and tracking, AI-based damage analysis, OCR-assisted document processing, machine-learning fraud detection, complaint handling, and administrative control within a single centralized platform. This gives users a faster and more transparent claim process and gives insurers better support for detecting fraud, while keeping data access controlled.

CHAPTER 2

REQUIREMENTS AND ANALYSIS

2.1 Existing System and Problem Statement

Existing System and Problem Statement Insurance claims are commonly handled through a combination of paper forms, emails, phone calls and basic office software. In a typical process, the policyholder fills in a claim form, attaches supporting documents such as identity proofs, bills, repair estimates and photographs of the damaged vehicle, and submits them to the insurance company. The company staff then examine the documents manually, verify the details provided, assess the damage and decide whether the claim should be approved or rejected. Claim records, complaints and customer communication are often maintained in separate files or applications. Although these methods provide the basic functions required for claim processing, they do not offer a unified environment for managing the complete claim process. Manual claim handling can make the process slow and inconvenient for both policyholders and insurance companies. Policyholders may have to visit offices or contact the company repeatedly to find out the status of their claims. Company staff may have to open and compare many documents and images before reaching a decision. Damage assessment normally depends on physical inspection or manual review of photographs, which can take considerable time and may vary from one reviewer to another. Similarly, general-purpose office tools and email do not normally include features such as structured claim tracking, automated damage analysis, document text extraction or claim-specific reporting. Another challenge is the detection of fraudulent claims. Fraud checking is generally carried out by reviewing submitted documents and claim details manually, which depends heavily on the experience and attention of the reviewing staff. Inconsistent information, altered documents or suspicious claim patterns may be difficult to identify when a large number of claims are received. The absence of an automated screening mechanism may make the verification process less consistent and may increase the risk of fraudulent claims being approved. Remote and digital claim submission also requires suitable mechanisms for managing users, documents and communication. In a typical manual system, there may be no dedicated way to approve registered users, store uploaded documents securely, record the verification decision with its reason, or handle enquiries and complaints in an organised manner. Even when such details are recorded, retrieving and reviewing them later can be difficult. This creates a need for a system that can manage claims, documents, users and communication in one place. These limitations indicate the need for a centralised platform that can bring claim submission, damage analysis, fraud detection, verification and communication together. The AI-Powered Smart Insurance Claim Management System is proposed to address this problem through a single web-based environment that supports

online claim submission, claim status tracking, AI-assisted vehicle damage analysis, OCR-based document processing, machine learning based fraud prediction, complaint and enquiry management and administrative control. By integrating these activities into one platform, the system aims to provide a more organised, efficient and reliable approach to managing insurance claims.

2.2 Solution Overview

The AI-Powered Smart Insurance Claim Management System is proposed as a unified web-based platform designed to simplify and improve the process of submitting, assessing and verifying insurance claims. Instead of depending on separate forms, emails and manual document checks, the system brings these activities together within a single environment. The platform provides dedicated access for three types of users, namely the User (policyholder), the Insurance Company and the Administrator, allowing each participant to manage claim-related activities through an organised workflow. The system allows users to register, log in and submit claims online by entering personal, policy, incident and vehicle details. Users can view the current status of their claims, view the insurance companies available on the platform, and send enquiries and complaints through the same interface. Insurance companies can register on the platform, view the claim requests received, examine the complete claim details and accept or reject claims. This reduces the need for physical visits and repeated follow-ups during the claim process. The system also provides features for AI-assisted vehicle damage analysis. After selecting an insurance company, the user can upload images of the vehicle from six sides, namely front, back, left, right, top and bottom. The insurance company can review these images along with the customer details and run the damage analysis, which generates an overall description of the damage. The status of the analysis is displayed to the user as pending, analyzing, completed or failed. This information can assist the company during claim assessment while keeping the final decision under the company's control. To improve the reliability of claim verification, the platform includes automated fraud detection features. Users can upload claim documents in formats such as PDF, DOCX, JPG and PNG. The system applies Optical Character Recognition (OCR) to extract text and relevant fields from the document, and a machine learning model predicts whether the claim is likely to be fraudulent along with a probability percentage. The insurance company can review the extracted data, the original document and the prediction, and then verify or reject the document with a note. The system records who verified the document and when, providing an additional layer of control over claim authenticity. The platform also supports user approval, complaint handling and administrative monitoring. The administrator can view registered insurance companies and users, approve or reject user registrations, and respond to complaints raised by users and companies. A dashboard presents summary information such as the number of companies, users and complaints through statistical cards and charts. By combining these functions

into one platform, the system provides a more structured, convenient and efficient approach to managing the complete insurance claim process.

2.3 Feasibility Study

The feasibility study is conducted to determine whether the proposed AI-Powered Smart Insurance Claim Management System can be developed and used successfully with the available technologies, resources and time. It examines the technical requirements, expected cost and practical usability of the system. Since the system combines claim management, damage analysis, OCR-based document processing, fraud prediction, verification and administration, checking its feasibility helps ensure that the proposed solution can be implemented effectively.

2.3.1 Technical Feasibility

The system is technically feasible because its major functions can be developed using commonly available web development and machine learning technologies. The web application can be built using the Django framework with a MySQL database for storing users, claims, documents and verification records. Text extraction from uploaded documents can be performed using EasyOCR, and fraud prediction can be implemented using machine learning models developed with Scikit-learn. Development and testing can be carried out using standard computers with an internet connection. The required software tools, libraries and frameworks are readily available, making the technical requirements manageable for the project.

2.3.2 Economic Feasibility

The proposed system is economically feasible because it can be developed without requiring expensive specialised hardware. Most of the development work can be carried out using open-source frameworks and libraries such as Django, MySQL, Scikit-learn, EasyOCR and Bootstrap. The system also reduces the paperwork, office visits and manual effort involved in handling claims. Although additional resources or external services may involve costs during large-scale deployment, the basic development and testing of the system can be completed with limited financial resources.

2.3.3 Operational Feasibility

The system is operationally feasible because it provides a simple and organised workflow for all its users. Insurance Companies can manage claim requests, review damage analysis and fraud detection results, and verify or reject documents through the platform, while Users can submit claims, upload images and documents, track claim status and send enquiries or complaints in the same environment.

Administrators can approve users and monitor the overall activity of the system. The centralised design reduces unnecessary manual work and makes claim information easier to manage. Since the system is web-based, users can access it through a supported browser without installing a separate desktop application. Therefore, the proposed system can be practically used for submitting, assessing and managing insurance claims.

CHAPTER 3
LITERATURE REVIEW

A literature survey was carried out to understand how earlier researchers have approached the individual capabilities that the AI-Powered Smart Insurance Claim Management System combines: prediction and assessment of insurance claims, detection of fraudulent claims using machine learning and deep learning, computer-vision-based vehicle damage assessment, and the handling of claims as a complete process. The surveyed works are discussed below with their techniques, contributions, limitations and relevance to the proposed system.

3.1 Fraud Detection and Analysis for Insurance Claim Using Machine Learning (2022)

Abhijeet Urunkar, Amruta Khot, Rashmi Bhat and Nandinee Mudegol presented a machine-learning-based approach for identifying fraudulent insurance claims. The study demonstrates how conventional machine-learning classifiers can be applied to insurance claim information to automatically distinguish potentially fraudulent claims from legitimate claims. The work investigates classifiers including Random Forest, Decision Tree and Support Vector Machine, showing the applicability of supervised machine learning to the insurance-fraud domain. This approach is important because it demonstrates that fraud screening does not necessarily require highly complex deep-learning architectures and can instead be performed using established machine-learning algorithms.

Technique: Random Forest, Decision Tree and Support Vector Machine classifiers are applied to insurance claim information.

Contribution: The study demonstrates the possibility of automatically analysing insurance claims and identifying suspicious claims using conventional machine-learning techniques.

Limitation: The study concentrates on fraud identification and does not integrate OCR for extracting information from uploaded insurance documents. It also does not provide vehicle-damage image analysis or a complete claim-management workflow.

Relevance: This work provides direct support for using classical machine-learning classifiers in the proposed system. The proposed system extends this concept by integrating fraud prediction with OCR-based document processing, company verification, AI-assisted vehicle-damage analysis and claim tracking. Paper: IEEE DOI – Fraud Detection and Analysis for Insurance Claim Using Machine Learning

3.2 Insurance Claims Estimation and Fraud Detection with Optimized Deep Learning Techniques (2025)

P. Anand Kumar and S. Sountharrajan proposed a deep-learning-based approach for insurance claim estimation and fraud detection. The study recognises that insurance companies need to estimate claim values accurately while simultaneously identifying potentially fraudulent claims. The researchers investigate several deep-learning architectures, including VGG16, VGG19, ResNet50 and custom convolutional neural networks. An Enhanced Hippopotamus Optimization Algorithm is combined with a custom 12-layer CNN to optimise model parameters and improve the performance of claim estimation and fraud detection. The reported results show that the EHOA-CNN-12 approach achieved approximately 92% accuracy, demonstrating the potential of optimised deep learning for complex insurance data.

Technique: Deep learning models including VGG16, VGG19, ResNet50 and CNN architectures are combined with an optimisation algorithm for insurance claim estimation and fraud detection.

Contribution: The major contribution is the integration of claim estimation and fraud detection into a deep-learning framework. The optimisation component attempts to improve the model's learning process and prediction performance.

Limitation: Deep-learning architectures and optimisation techniques can require substantial computational resources, larger training datasets and greater implementation complexity. Such models can therefore be difficult to deploy in lightweight academic or small-organisation environments.

Relevance: The study confirms that claim estimation and fraud detection can be addressed together, but the proposed system chooses comparatively lightweight Scikit-learn-based machine-learning techniques so that fraud prediction can be integrated into a Django web application running on standard computing resources. Paper: Scientific Reports – Insurance claims estimation and fraud detection with optimized deep learning techniques

3.3 Machine Learning Based Method for Insurance Fraud Detection on Class Imbalance Datasets With Missing Values (2024)

Ahmed A. Khalil, Zaiming Liu, Ahmed Fathalla, Ahmed Ali and Ahmad Salah investigate the problem of insurance fraud detection when datasets contain both missing values and severe class imbalance. This is particularly important because fraudulent insurance claims generally form a much smaller proportion of the overall claim population than legitimate claims. Consequently, a conventional machine-learning classifier may become biased towards the majority class and fail to

identify fraudulent cases effectively. The study investigates data preparation, missing-value handling, imbalanced-data processing and different machine-learning methodologies using insurance-fraud datasets. The researchers also evaluate the performance of their approaches using a public automobile insurance dataset and compare the resulting models with existing approaches.

Technique: Machine learning is combined with data cleaning, missing-value treatment, feature analysis and resampling strategies for handling class-imbalanced insurance datasets.

Contribution: The study highlights the importance of preprocessing and class-imbalance handling before training an insurance fraud-detection model. It demonstrates that simply applying a classifier to raw insurance data may not be sufficient for reliable fraud identification.

Limitation: The research is primarily focused on the fraud-detection modelling problem. It does not provide a complete claim-management platform containing claim submission, OCR document extraction, company verification, vehicle-damage analysis and claim-status tracking.

Relevance: This work directly supports the proposed system's requirement to prepare insurance information appropriately before fraud prediction. It also reinforces the importance of considering the imbalance between genuine and fraudulent claims when developing the Decision Tree-based fraud-detection component. Paper: IEEE DOI – Machine Learning Based Method for Insurance Fraud Detection on Class Imbalance Datasets With Missing Values

3.4 Insurance Claim Prediction and Risk Assessment Using Machine Learning (2025)

This work investigates the application of machine learning to insurance claim prediction and customer risk assessment. The study addresses the limitations of traditional insurance claim evaluation, where large volumes of historical claim information can make manual analysis expensive, time-consuming and susceptible to human error. The researchers propose a machine-learning-based approach that uses historical insurance data to predict claim outcomes and assess the associated risk. The study evaluates Ridge Regression and Logistic Regression for categorising claim outcomes and reports that Ridge Regression achieved better prediction performance than Logistic Regression. The results demonstrate that machine learning can support insurers in making more consistent and data-driven decisions when evaluating claims and customer risk.

Technique: Regression and classification-based machine learning models are trained using historical insurance claim information. Ridge Regression and Logistic Regression are used to predict and categorise claim outcomes.

Contribution: The study demonstrates that machine learning can improve insurance claim prediction and risk assessment. It provides evidence that historical claim information can be transformed into

useful predictive information, helping insurance organisations make faster and more consistent decisions.

Limitation: The work mainly focuses on claim prediction and risk assessment. It does not provide an integrated fraud-detection mechanism, OCR-based document processing, vehicle-damage image analysis, or a complete claim-management workflow.

Relevance: This study supports the use of machine learning for insurance decision support. The proposed AI-Powered Smart Insurance Claim Management System extends this concept by combining machine-learning-based fraud prediction with document processing, vehicle-damage analysis, claim verification and end-to-end claim tracking. Paper: IEEE Xplore – Insurance Claim Prediction and Risk Assessment Using Machine Learning

3.5 Insurance Fraud Detection: Evidence from Artificial Intelligence and Machine Learning (2022)

Faheem Aslam, Ahmed Imran Hunjra, Zied Ftiti, Wael Louhichi and Tahira Shams investigated artificial-intelligence and machine-learning approaches for detecting fraud in the automobile insurance sector. The study compares different predictive models and investigates which variables are most influential for identifying fraudulent claims. Logistic Regression, Support Vector Machine and Naïve Bayes are considered, while Boruta is used for feature selection. The research reports that machine-learning algorithms can improve fraud-detection capability and that feature selection can help identify variables that contribute significantly to the prediction process. The study also reports that SVM performs strongly in terms of accuracy while Logistic Regression achieves a high F-measure.

Technique: Logistic Regression, Support Vector Machine, Naïve Bayes and Boruta-based feature selection.

Contribution: The study contributes an empirical comparison of machine-learning algorithms and investigates the importance of different features for automobile insurance fraud detection.

Limitation: The study concentrates on predictive fraud detection and does not address OCR-based insurance document extraction, vehicle-image damage assessment, user dashboards or complete claim workflow management.

Relevance: The comparison of machine-learning algorithms and identification of fraud-related features provide useful guidance for selecting an appropriate fraud model. The proposed system builds upon this model-centred approach by placing fraud prediction inside a complete web-based

claim-management platform. Paper: ScienceDirect – Insurance fraud detection: Evidence from artificial intelligence and machine learning

3.6 Stacking Classifier-Based Automated Insurance Fraud Detection System (2022)

M. Shanthini proposed a stacking-classifier-based automated insurance fraud-detection system that combines machine-learning classification with a web-based insurance workflow. The study is particularly relevant because it does not consider fraud detection as an isolated prediction problem; instead, it also considers the digitalisation of insurance processes and documentation. The proposed framework uses a stacking classifier and addresses data-management issues through linear-regression-based iterative imputation. The work aims to improve the efficiency of paper-based insurance processes by integrating machine learning with a web application for automated fraud detection and claim processing.

Technique: Stacking ensemble classification with data imputation and a web-based insurance-processing environment.

Contribution: The study demonstrates how machine learning can be integrated with a web application to automate fraud screening and improve the management of insurance information.

Limitation: Although it addresses digital claim processing, it does not provide the same combination of OCR-based document extraction, AI vehicle-damage analysis and detailed user/company/admin claim tracking proposed in the current system.

Relevance: This work is especially relevant to the proposed system because both approaches combine machine-learning-based fraud detection with a web-based insurance workflow. The proposed system further extends the concept through OCR, image-based vehicle-damage analysis, claim verification and status tracking. Paper: IEEE DOI – Stacking Classifier-Based Automated Insurance Fraud Detection System

3.7 Detecting Insurance Fraud Using Supervised and Unsupervised Machine Learning (2023)

Jörn Debener, Volker Heinke and Johannes Kriebel investigated insurance fraud detection using both supervised and unsupervised machine-learning techniques. The research is significant because insurance fraud datasets commonly contain relatively few labelled fraudulent claims, making it difficult to rely entirely on supervised learning. The authors therefore compare supervised approaches with unsupervised anomaly-detection methods, including isolation forests, using proprietary insurance claim data. They additionally conduct a field experiment with an insurance company to evaluate the practical ability of the models to identify new fraudulent claims. The results indicate that

unsupervised learning, particularly isolation forests, can detect fraudulent claims, while supervised learning can also perform strongly even with relatively few labelled fraud cases. The authors consequently argue that supervised and unsupervised approaches should be treated as complementary rather than mutually exclusive.

Technique: Supervised machine learning and unsupervised anomaly detection, including isolation forests.

Contribution: The study provides a comparison between two major machine-learning strategies for insurance fraud detection and demonstrates that both can provide useful information for fraud screening.

Limitation: The research uses proprietary insurance data and focuses specifically on fraud detection. It does not provide an integrated web-based claim-management system containing OCR, vehicle-image analysis, document verification and claim-status tracking.

Relevance: The study supports the use of machine learning as an assisted fraud-screening mechanism and demonstrates the potential value of combining different fraud-detection strategies. The proposed system currently uses a supervised Decision Tree classifier as a practical and interpretable decision-support mechanism that can be integrated into the Django application. Paper: Wiley Online Library – Detecting insurance fraud using supervised and unsupervised machine learning

3.8 Automobile Insurance Fraud Detection Using Data Mining: A Systematic Literature Review (2024)

Gilian Schrijver, Dipti K. Sarmah and Mohammed El-hajj conducted a systematic literature review of recent research on automobile insurance fraud detection. The review examines 50 publications covering research published between January 2019 and March 2023 and analyses the datasets, resampling methods and data-mining techniques used in insurance-fraud detection. The review found that supervised machine-learning methods using structured claim and policy information are particularly common. Random Forest, Logistic Regression, SVM, Decision Tree and XGBoost are among the frequently used techniques. The authors also identify alternative research directions involving unsupervised learning, natural-language processing, graph-based approaches and cost-sensitive classification.

Technique: Systematic literature review covering machine learning, data mining, resampling and fraud-detection approaches.

Contribution: The study provides a broad overview of the insurance-fraud research landscape and identifies commonly used datasets, preprocessing methods, supervised approaches and alternative research directions.

Limitation: The review is limited to the selected publication period and depends on the quality and availability of the studies included. It does not itself implement a complete insurance claim-management system.

Relevance: This systematic review strongly supports the proposed system's use of structured insurance information and classical machine-learning algorithms for fraud detection. It also identifies the potential value of incorporating unstructured text and image information in future versions, which is consistent with the proposed system's OCR and vehicle-damage-analysis modules. Paper: ScienceDirect – Automobile insurance fraud detection using data mining: A systematic literature review

3.9 Computer Vision Based Automatic Car Damage Analysis (2023)

The CDA-Net: Computer Vision based Automatic Car Damage Analysis study investigates the automation of vehicle-damage assessment using computer vision and deep-learning techniques. The study recognises that conventional automobile insurance claims frequently require manual inspection by engineers and surveyors, which can take several days or even weeks. To address this problem, the researchers developed datasets containing vehicle make/model information and annotated vehicle-damage images. The damage dataset contains thousands of images representing different types of vehicle damage, including dents, scratches, bends, broken components, cracks and other damage categories. The study also considers the damaged component, damage type, severity and location, demonstrating how image-based analysis can provide useful information for automated insurance claim assessment.

Technique: Computer vision and deep-learning-based vehicle image analysis.

Contribution: The work demonstrates how vehicle-damage images can be analysed automatically to identify damaged components, damage categories, severity and locations. It provides a foundation for replacing or assisting manual vehicle inspection.

Limitation: The research primarily concentrates on computer-vision-based damage analysis rather than providing a complete insurance claim-management workflow. It does not integrate claim submission, fraud-document OCR, company verification and claim-status tracking into one platform.

Relevance: This study is directly relevant to the proposed AI-assisted vehicle-damage-analysis component. The proposed system extends image-based damage assessment by integrating multiple

uploaded vehicle views into a broader insurance claim workflow, allowing the insurance company to review the analysis as part of claim processing. Paper: ACM Digital Library – CDA-Net: Computer Vision based Automatic Car Damage Analysis

3.10 A Survey on Vehicle Damage Detection Using Deep Learning Towards Intelligent Insurance (2025)

Doan-Hieu Tran, Van-Dung Hoang, Dung Nguyen, Thanh-An Pham and Van-Tuong-Lan Le presented a survey and experimental comparison of modern deep-learning object-detection approaches for vehicle-damage detection in intelligent insurance applications. The study explains that vehicle-damage identification is important for determining damage severity and estimating repair costs, both of which are important parts of automobile insurance claim processing. The researchers compare models including RT-DETR, YOLOv12n, YOLOv11n and YOLOv8n and investigate their ability to identify small and complex damage areas under different lighting and viewing conditions. Their reported results show differences in detection performance among the models, with RT-DETR achieving the highest reported mAP in their evaluation.

Technique: Deep-learning-based object detection using modern YOLO and transformer-based detection architectures.

Contribution: The study demonstrates the potential of automated object detection for identifying vehicle damage and supporting more efficient insurance claim assessment.

Limitation: The study concentrates on image-based vehicle-damage detection and does not provide a complete claim-management platform incorporating document OCR, fraud prediction, user dashboards and insurance-company verification.

Relevance: The research supports the proposed use of AI-based vehicle-image analysis in insurance claims. It also indicates that image-based damage detection can reduce dependence on completely manual inspection and potentially improve the speed and consistency of claim assessment. Paper: IEEE DOI – A Survey on Vehicle Damage Detection using Deep Learning Towards Intelligent Insurance

3.11 Summary of the Literature Survey

Table 3.1: Summary of the surveyed literature

Work (Year)	Authors	Technique	Contribution	Limitation
Fraud Detection and Analysis for Insurance Claim Using Machine Learning (2022)	Abhijeet Urunkar et al.	Random Forest, Decision Tree, SVM	Automatically identifies potentially fraudulent claims	No OCR, vehicle-damage analysis or complete workflow
Insurance Claims Estimation and Fraud Detection with Optimized Deep Learning Techniques (2025)	P. Anand Kumar, S. Sountharrajan	VGG16, VGG19, ResNet50, CNN, EHOA	Combines claim estimation and fraud detection	High computational requirements; complex implementation
ML Based Method for Insurance Fraud Detection on Class Imbalance Datasets With Missing Values (2024)	Ahmed A. Khalil, Zaiming Liu, Ahmed Fathalla, Ahmed Ali, Ahmad Salah	ML, preprocessing, missing-value handling, resampling	Improves fraud detection with imbalanced and incomplete data	Mainly fraud detection; lacks full claim management
Insurance Claim Prediction and Risk Assessment Using ML (2025)	Various authors	Ridge Regression, Logistic Regression	Predicts claim outcomes and assesses customer risk	No automated fraud detection, OCR or complete claim processing
Insurance Fraud Detection: Evidence from AI and ML (2022)	Faheem Aslam et al.	Logistic Regression, SVM, Naïve Bayes, Boruta	Compares ML algorithms and identifies fraud-related features	No OCR, dashboard or complete claim management

Stacking Classifier-Based Automated Insurance Fraud Detection System (2022)	M. Shanthini	Stacking classifier, data imputation	Automates fraud screening within a web-based insurance process	No integrated OCR and vehicle-damage analysis
Detecting Insurance Fraud Using Supervised and Unsupervised ML (2023)	Jörn Debener, Volker Heinke, Johannes Kriebel	Supervised ML, unsupervised ML, Isolation Forest	Compares supervised and anomaly-detection approaches	Proprietary data; focuses mainly on fraud detection
Automobile Insurance Fraud Detection Using Data Mining: A Systematic Literature Review (2024)	Gilian Schrijver, Dipti K. Sarmah, Mohammed El-hajj	Systematic review, ML, data mining, resampling	Reviews 50 studies and identifies research directions	Limited period/public datasets; no complete implementation
CDA-Net: Computer Vision Based Automatic Car Damage Analysis (2023)	CDA-Net authors	Computer vision, deep learning	Detects vehicle damage, type, severity and location	Focuses on image analysis rather than complete insurance workflow
A Survey on Vehicle Damage Detection Using Deep Learning Towards Intelligent Insurance (2025)	Doan-Hieu Tran et al.	RT-DETR, YOLOv12n, YOLOv11n, YOLOv8n	Evaluates modern models for automated vehicle-damage detection	No integrated fraud, OCR or claim-management workflow

3.12 Research Gap Identified

• Most works concentrate on one task, such as claim prediction, fraud detection or vehicle-damage detection, and do not provide a complete workflow that covers claim submission, review, decision and tracking.

• The surveyed works do not report the use of OCR to extract information from the documents that policyholders submit, so fraud checking is not directly linked to the claim documents themselves.

• Deep-learning approaches for fraud or image analysis can require high computational resources and are comparatively complex to implement. A lighter approach based on classical machine learning has received less attention in complete claim-management systems.

• Little of the surveyed work offers dashboards and claim tracking for users, insurance companies and administrators within one platform.

• Few solutions combine open-source technologies such as Python, Django, MySQL, Scikit-learn and OCR/image-analysis tools into a deployable academic claim-management platform.

The AI-Powered Smart Insurance Claim Management System addresses these gaps by combining online claim submission, OCR-based document processing, machine-learning-based fraud prediction with a risk probability, AI-assisted vehicle damage analysis, a claim approval and rejection workflow, real-time claim tracking and an administrator dashboard in one web-based platform.

CHAPTER 4
DIAGRAMS

4.1 Data Flow Diagrams (DFD)

A Data Flow Diagram (DFD) is a graphical representation of how data move through a system. It shows the external entities that supply or receive data, the processes that transform the data, and the data stores in which data are kept. A DFD can be drawn at different levels of detail. The context-level (Level 0) diagram shows the system as a single process, and the Level 1 diagrams break that process into its main sub-processes. The Data Flow Diagrams (DFD) illustrate how data moves between the **User, Insurance Company, Admin, system processes, and data stores** in the AI-Powered Smart Insurance Claim Management System.

4.1.1 Level 0 DFD (Context Diagram)

Figure 4.1 shows the context-level view of the system, representing the AI-Powered Smart Insurance Claim Management System as a single process. The User, Insurance Company, and Admin interact with the system by sending requests and receiving responses related to claims, verification, fraud detection, and system management.

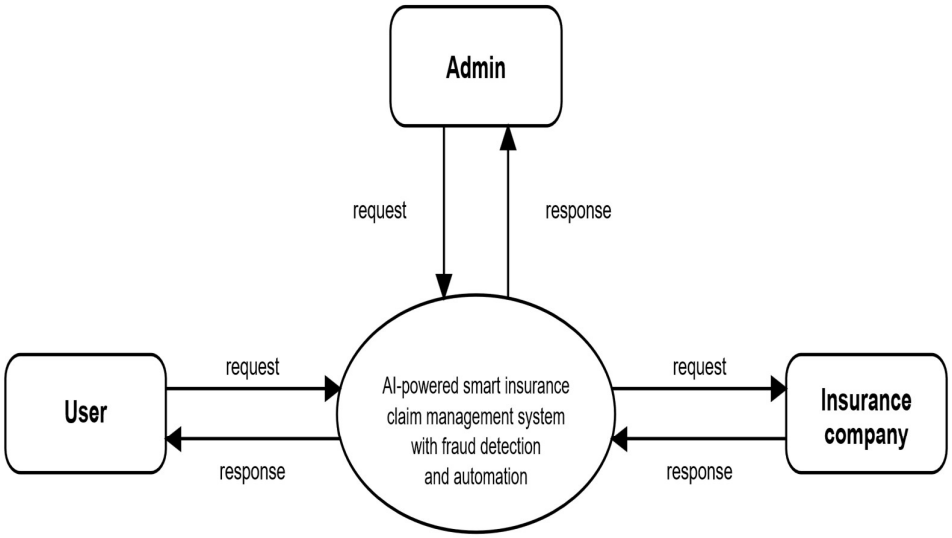


Figure 4.1: Level 0 DFD of the AI-Powered Smart Insurance Claim Management System

4.1.2 Level 1 DFD: User

Figure 4.2 describes the data flow from the User's perspective. The main processes include submitting claims, uploading documents, OCR processing, fraud detection, claim approval or rejection, receiving claim decisions and notifications, and tracking claim status

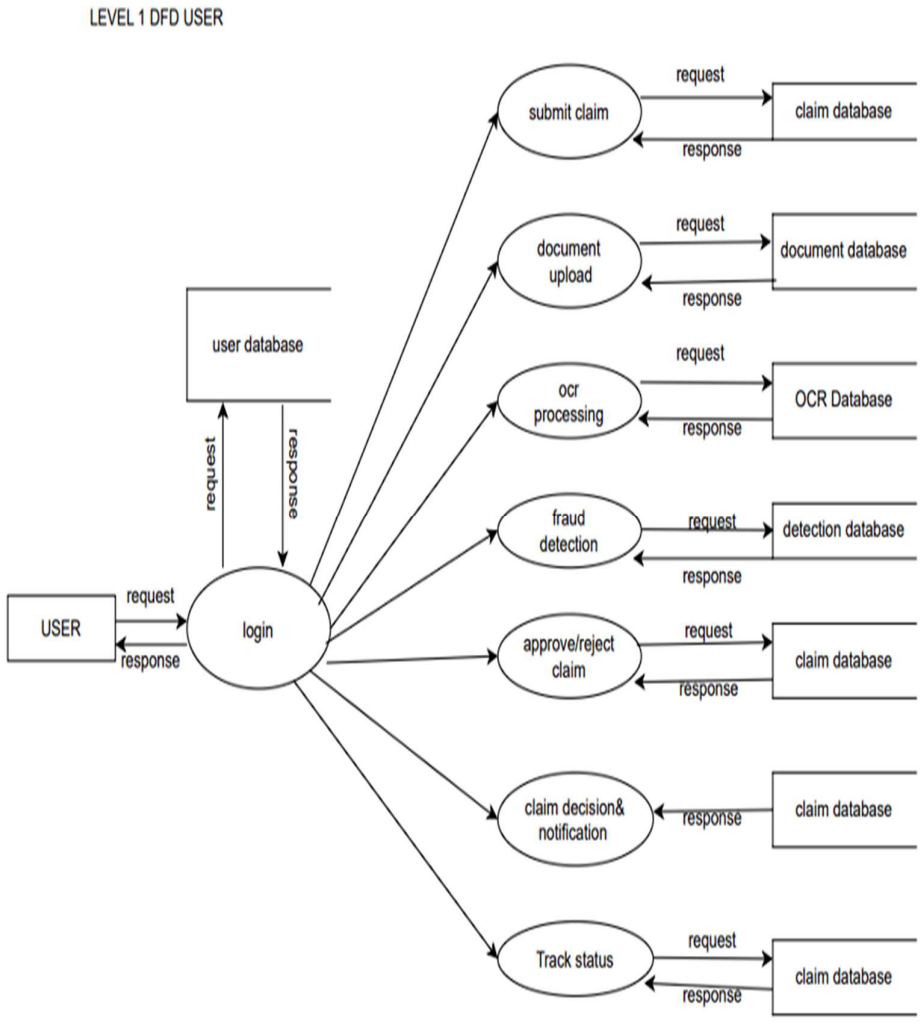


Figure 4.2: Level 1 DFD: User

4.1.3 Level 1 DFD: Insurance Company

Figure 4.3 represents the main activities performed by the Insurance Company. The processes include viewing claim requests, verifying documents, viewing OCR data, viewing fraud scores, approving or rejecting claims, sending notifications, and updating claim status using the relevant data stores.

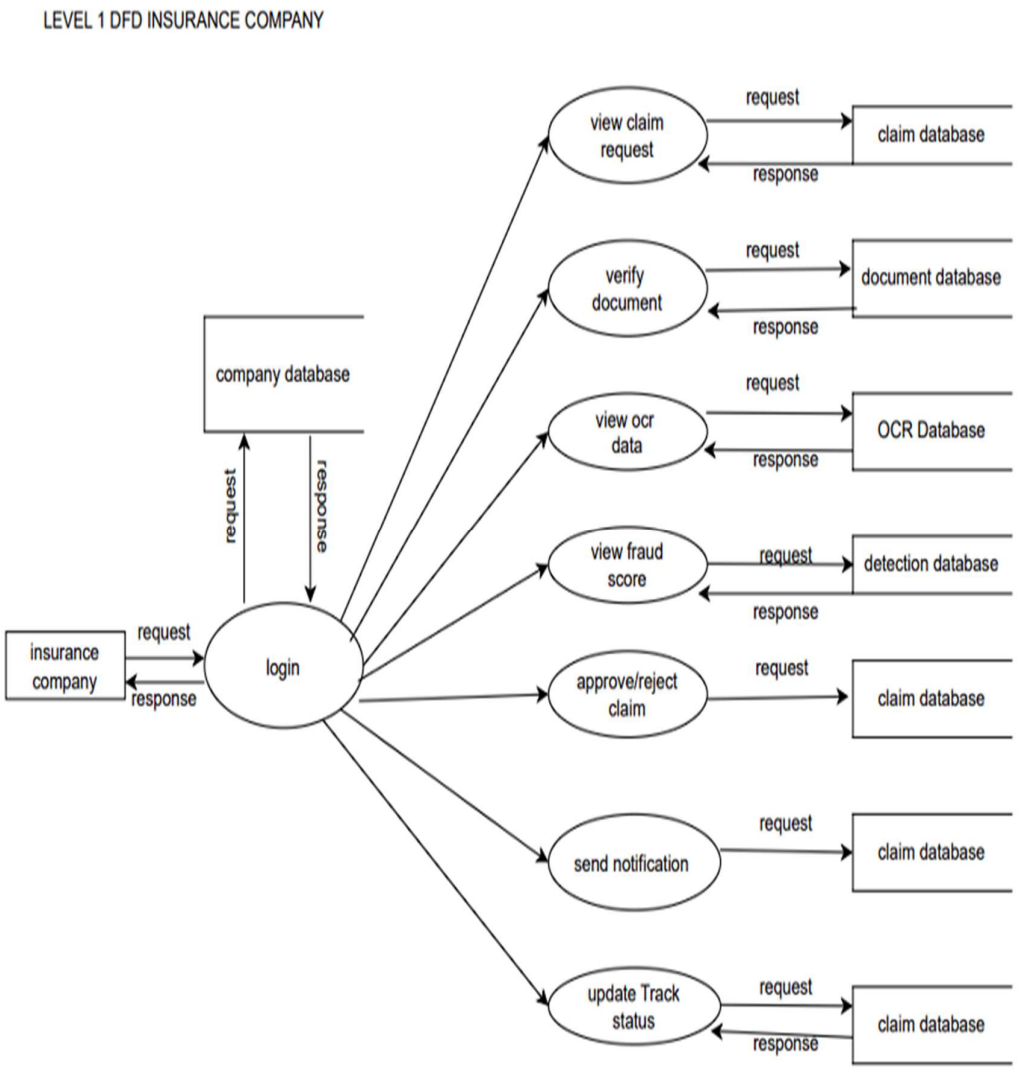


Figure 4.3: Level 1 DFD: Insurance Company

4.1.4 Level 1 DFD: Admin

Figure 4.4 expands the Admin's interaction with the system after login. It includes processes for managing users, managing insurance companies, monitoring claims, viewing fraud reports, and generating reports, with data exchanged between the processes and their respective databases.

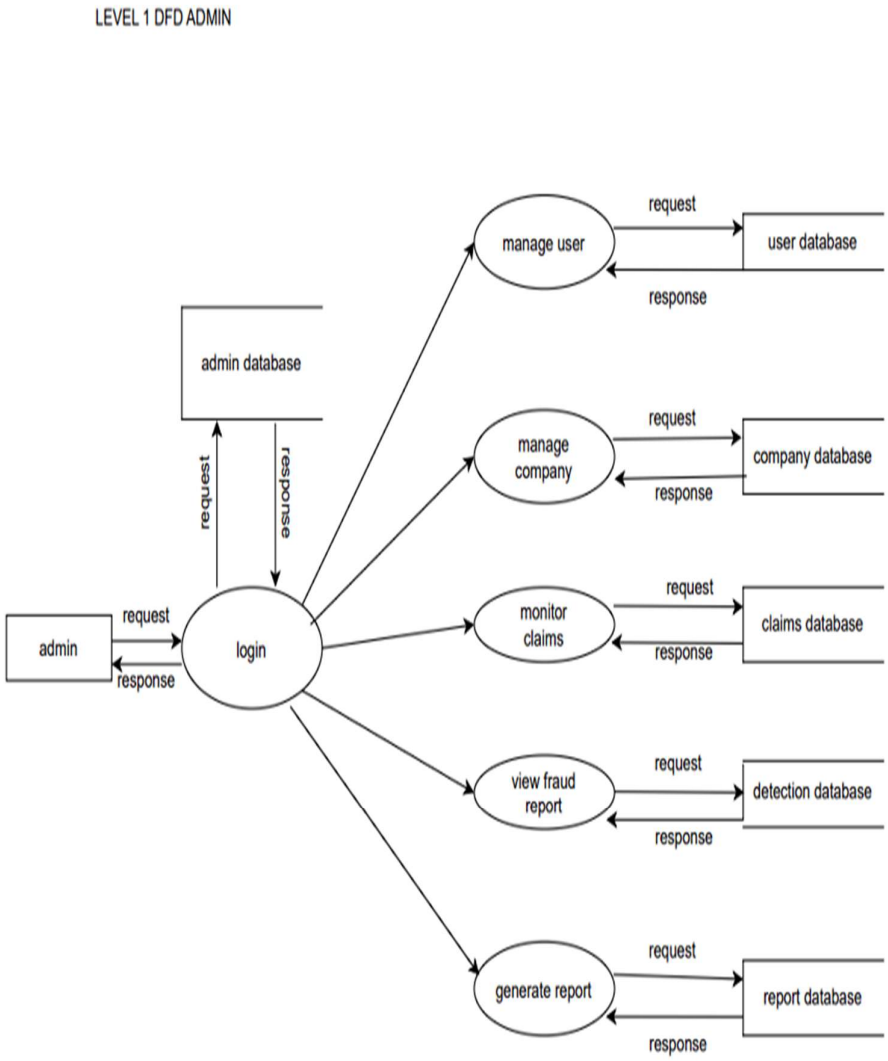


Figure 4.4: Level 1 DFD: Admin

4.2 Use Case Diagram

The use case diagram in Figure 4.5 describes the system from the users' point of view and identifies three main actors: the User, the Insurance Company and the Admin.

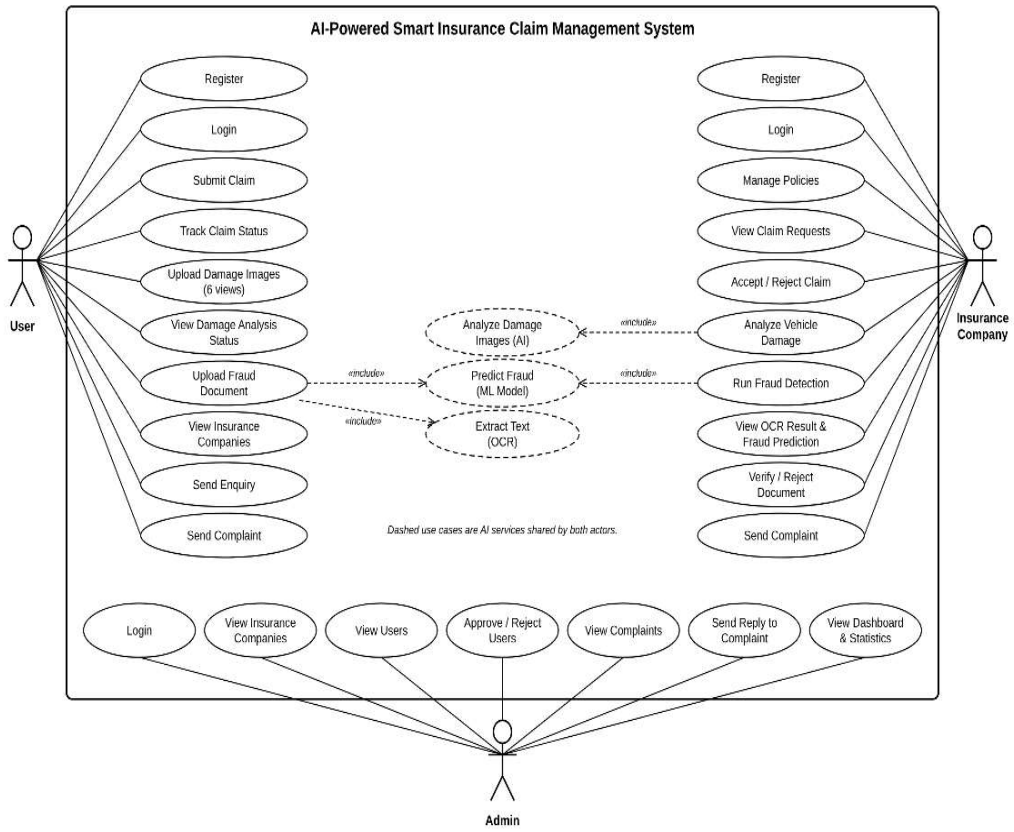


Figure 4.5: Use Case Diagram of the AI-Powered Smart Insurance Claim Management System

• User: Register, Login, Submit Claim, Track Claim Status, Upload Damage Images, View Damage Analysis, Upload Fraud Document, View Insurance Companies, Send Enquiry and Send Complaint.

• Insurance Company: Register, Login, Manage Policies, View Claim Requests, Analyze Vehicle Damage, Run Fraud Detection, View OCR and Fraud Prediction Results, Verify/Reject Documents, Accept/Reject Claims and Send Complaint.

• Admin: Login, View Insurance Companies, View Users, Approve/Reject Users, View Complaints, Reply to Complaints and View Dashboard and Statistics

4.3 Class Diagram

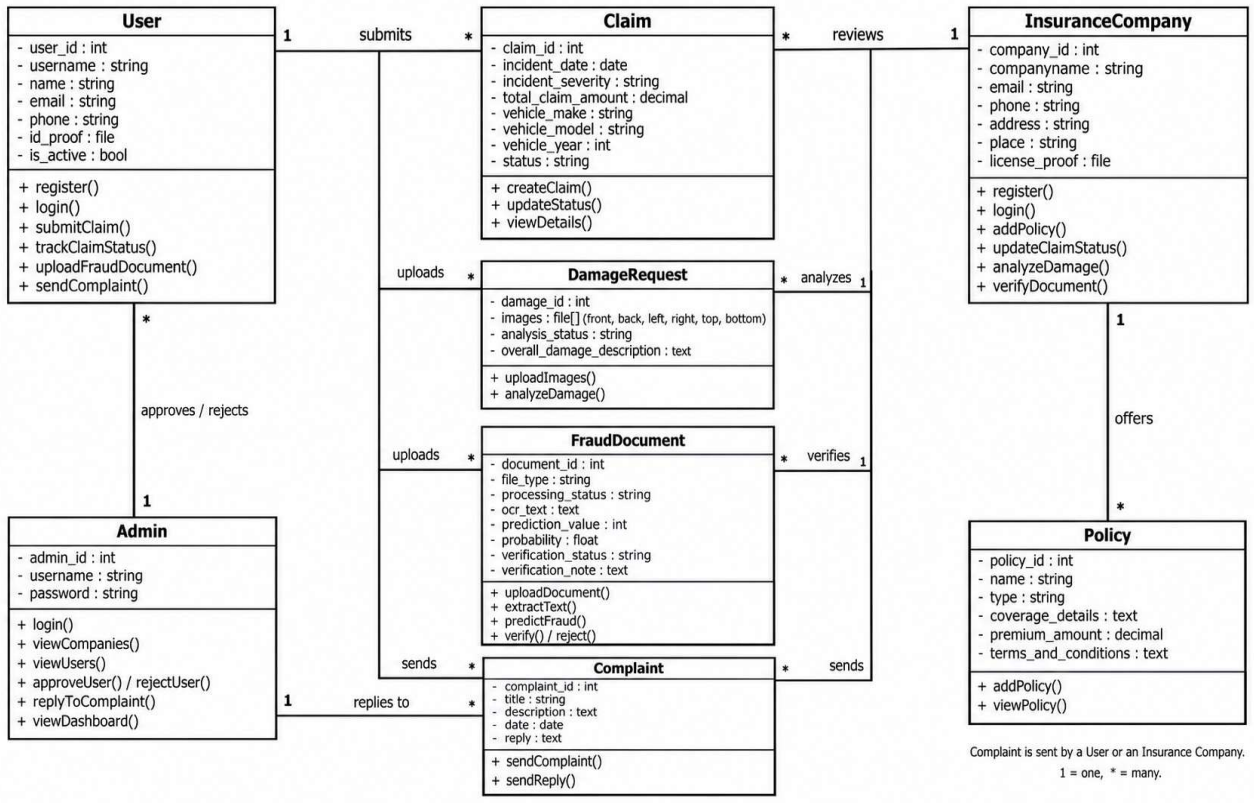
The class diagram in Figure 4.6 represents the static structure of the AI-Powered Smart Insurance Claim Management System. It shows the main classes, their attributes, operations and relationships. The User class manages registration, login, claim submission, document uploading and claim tracking. The InsuranceCompany class manages policies, claims, damage analysis and document verification. The Claim class stores claim details and status, while DamageRequest manages vehicle images and AI-generated damage descriptions. The FraudDocument class handles document upload,

OCR extraction and fraud prediction. The Policy, Complaint and Admin classes support policy management, communication and administrative functions.

Figure 4.6: Class Diagram of the AI-Powered Smart Insurance Claim Management System

4.4 State Diagram

The state diagram in Figure 4.7 represents the life cycle of an insurance claim from initial submission



to the final decision. The process begins when the policyholder submits a claim, and the claim is assigned a **Pending** status. At this stage, the required claim details, supporting documents and vehicle images are uploaded to the system. The submitted information is then passed to the appropriate processing components for further analysis.

Once the claim is submitted, the system performs **OCR processing** on the uploaded documents to extract relevant information. At the same time, the fraud-detection module analyses the available claim data to identify possible fraudulent patterns, while the AI-based vehicle damage analysis examines the uploaded vehicle images to identify and assess visible damage. These automated processes reduce manual effort and provide supporting information for claim evaluation.

If any of the processing operations fail or produce an incomplete result, the corresponding operation can be **retried**. This allows temporary processing errors to be handled without restarting the entire

claim process. After the required OCR, fraud analysis and vehicle damage analysis are successfully completed, the claim moves to the **Awaiting Verification** state.

In the Awaiting Verification state, the Insurance Company reviews the submitted claim information together with the OCR results, fraud-prediction output and vehicle-damage analysis. Based on the available evidence and verification process, the insurance company makes the final decision. If the claim satisfies the required conditions, its status changes to **Approved**. If the claim is found to be invalid, suspicious or does not satisfy the required conditions, it changes to **Rejected**.

Finally, the updated claim status is stored in the system and displayed to the policyholder through the user portal. Thus, the state diagram demonstrates the complete transition of a claim from **Pending → Processing → Awaiting Verification → Approved/Rejected**, including retry handling for unsuccessful processing operations.

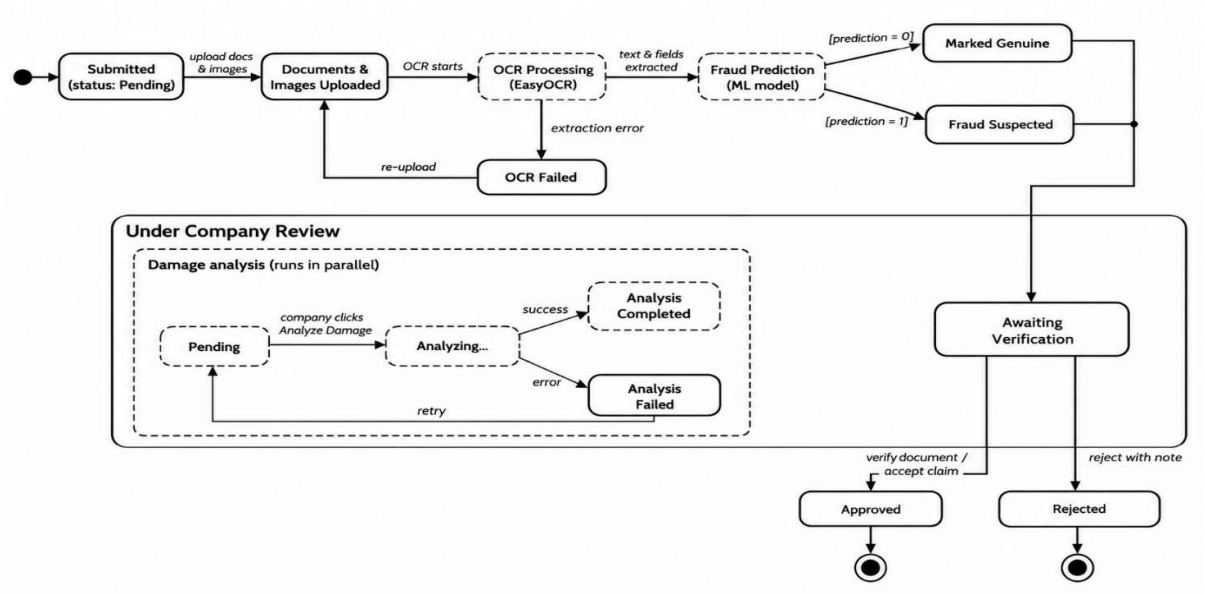


Figure 4.7: State Diagram of the AI-Powered Smart Insurance Claim Management System

4.5 Activity Diagram

The activity diagram in Figure 4.8 illustrates the workflow of the system using different activities performed by the User, System, OCR & ML Engine, Damage Analysis AI, and Insurance Company. The user logs in and submits claim details, documents and vehicle damage images. The system stores the claim and sends the uploaded information for OCR, fraud prediction and AI-based damage analysis. After processing, the Insurance Company reviews the generated results and verifies or rejects the claim. The system then updates the claim status and allows the user to view the final decision.

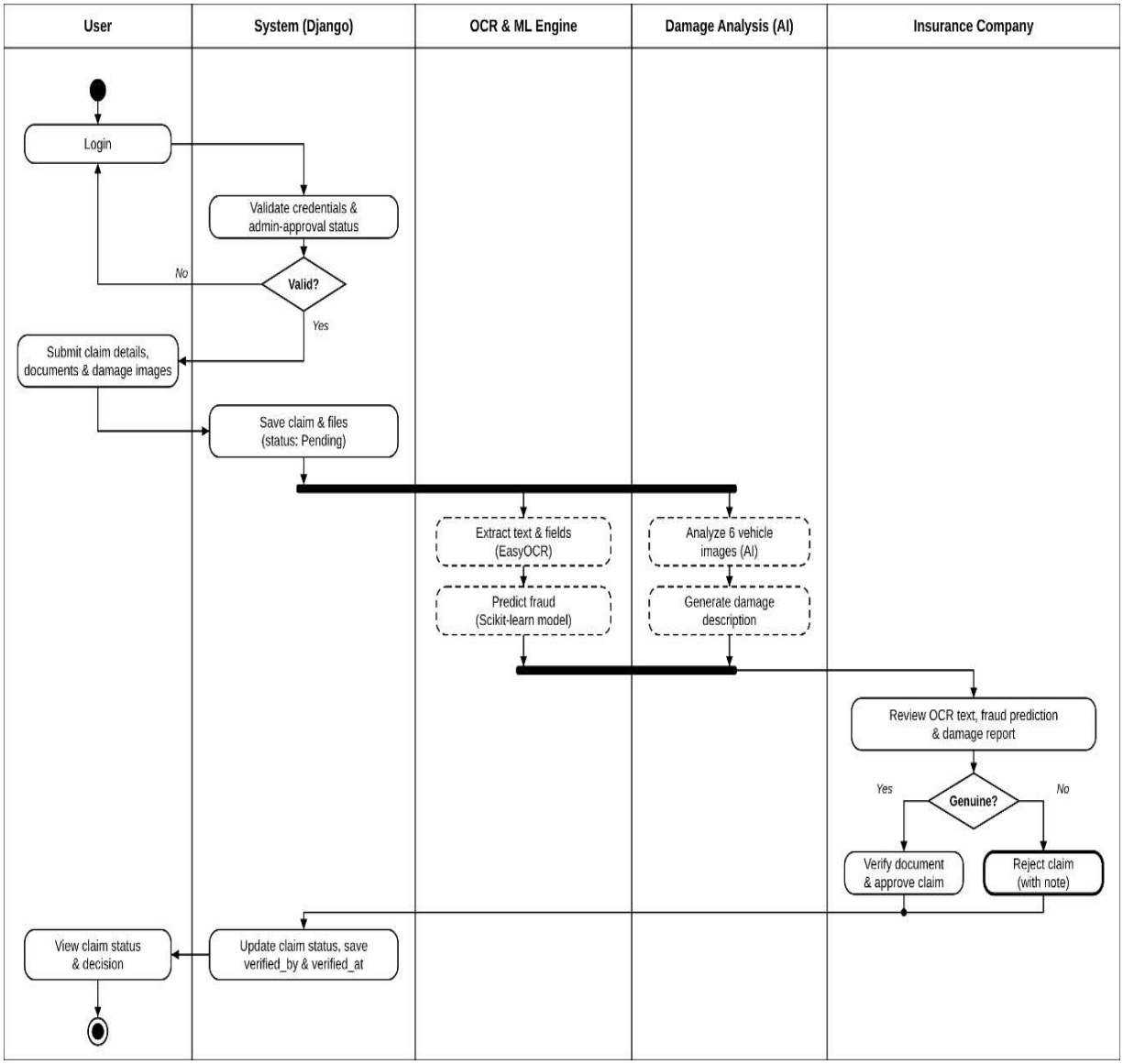


Figure 4.8: Activity Diagram of the AI-Powered Smart Insurance Claim Management System

4.6 Sequence Diagram

The sequence diagram in Figure 4.9 shows the chronological interaction between the Admin, User, Web Application, MySQL Database, OCR & ML Engine, Damage Analysis AI, and Insurance Company. The interaction can be divided into the following stages:

1. Registration: The user registers by providing details and ID proof. The system stores the registration and the Admin approves the account.
2. Login: The user logs in and the system validates the credentials using the database.
3. Claim Submission: The user submits claim details and uploads documents and vehicle damage images.

4. AI Processing: The OCR & ML Engine extracts document information and predicts fraud, while the Damage Analysis AI analyzes the vehicle images.
5. Company Review: The Insurance Company reviews the OCR results, fraud prediction and damage analysis report.
6. Decision: The company verifies or rejects the claim, and the system stores the decision and displays the final status to the user.

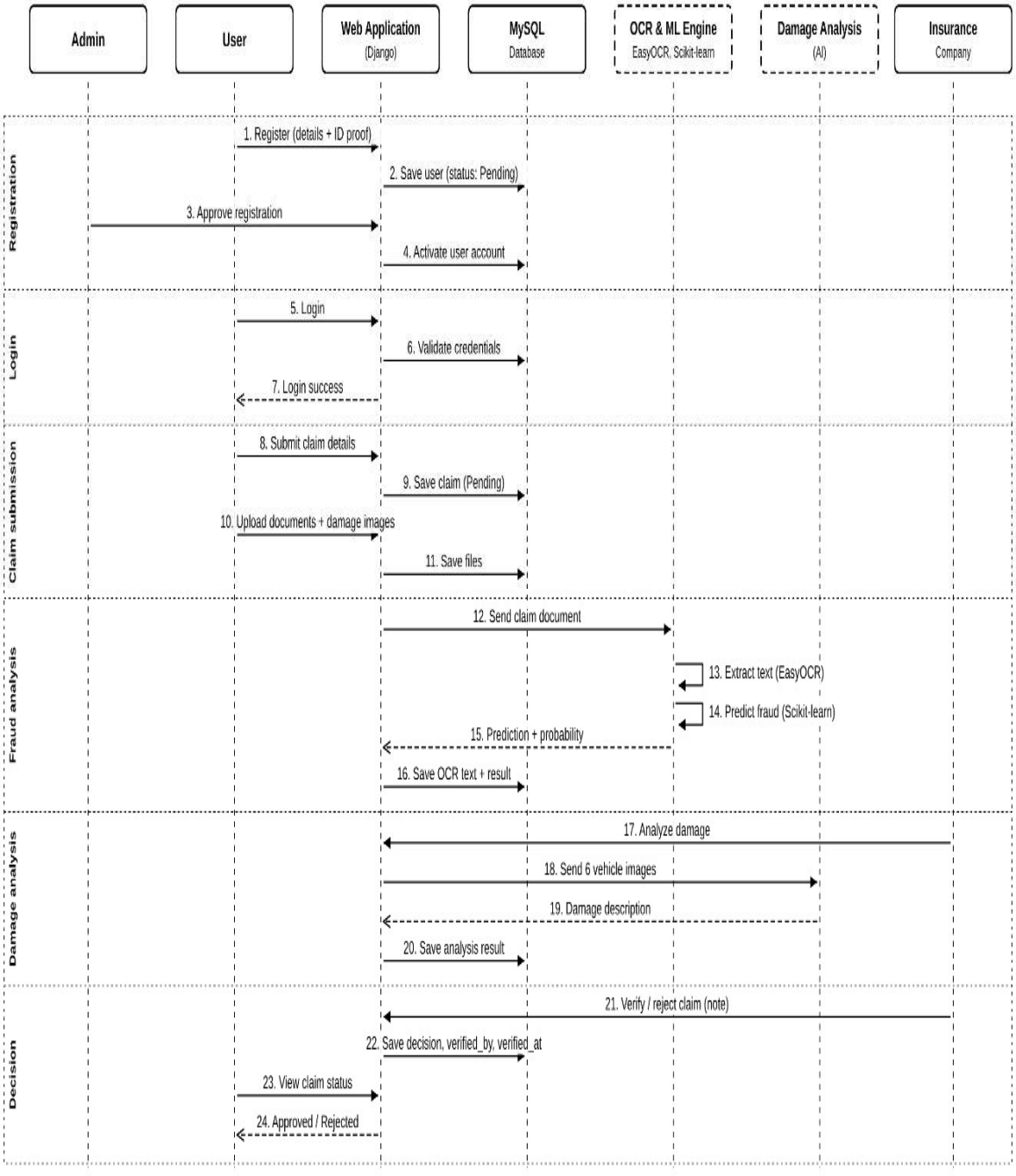


Figure 4.9: Sequence Diagram of the AI-Powered Smart Insurance Claim Management System

CHAPTER 5
METHODOLOGY

5.1 Module Descriptions

The AI-Powered Smart Insurance Claim Management System is divided into six major modules. Each module handles a specific set of functions and interacts with other modules to provide an automated and efficient insurance claim processing workflow.

No.	Module	Responsibility
1	User Module	Handles user registration, login, profile management and claim access.
2	Insurance Company Management Module	Manages insurance companies, policies, claim requests and claim verification
3	Claim Management Module	Handles claim submission, document upload, claim details and claim status tracking.
4	Fraud Detection and OCR Module	Extracts information from uploaded documents using OCR and predicts possible fraud using the ML model.
5	AI Vehicle Damage Analysis Module	Analyzes uploaded vehicle damage images using AI and generates a damage description.
6	Admin Management Module	Manages users, insurance companies, approvals, complaints and system monitoring.

Table 5.1: Modules of Insurance Claim Management System

5.1.1 User Module

The User Management Module manages customer-related activities. Users can register by providing their personal details and ID proof. Registered users can log in after account approval and access their dashboard. The module also allows users to view insurance companies, submit claims, upload required documents and vehicle images, and track the status of their claims.

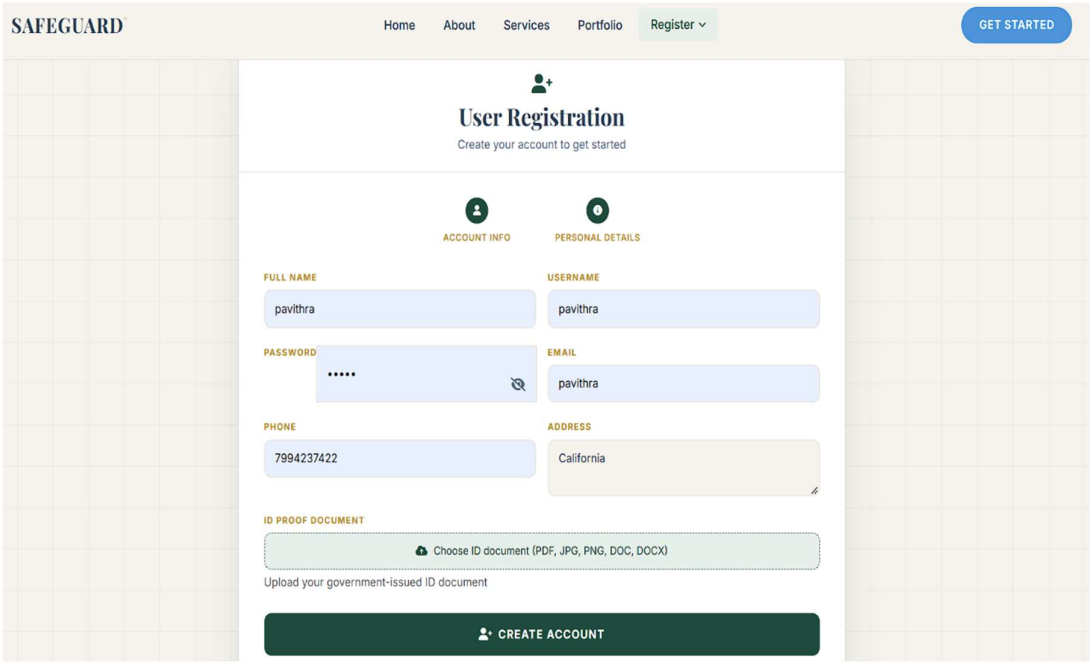


Figure 5.1: User Registration and Login Interfaces

5.1.2 Insurance Company Management Module

The Insurance Company Management Module provides functions for insurance companies to manage their activities. Companies can register and log in, manage insurance policies, view submitted claims, examine uploaded documents and review AI-generated fraud and damage analysis results. The company can verify documents and approve or reject claims.

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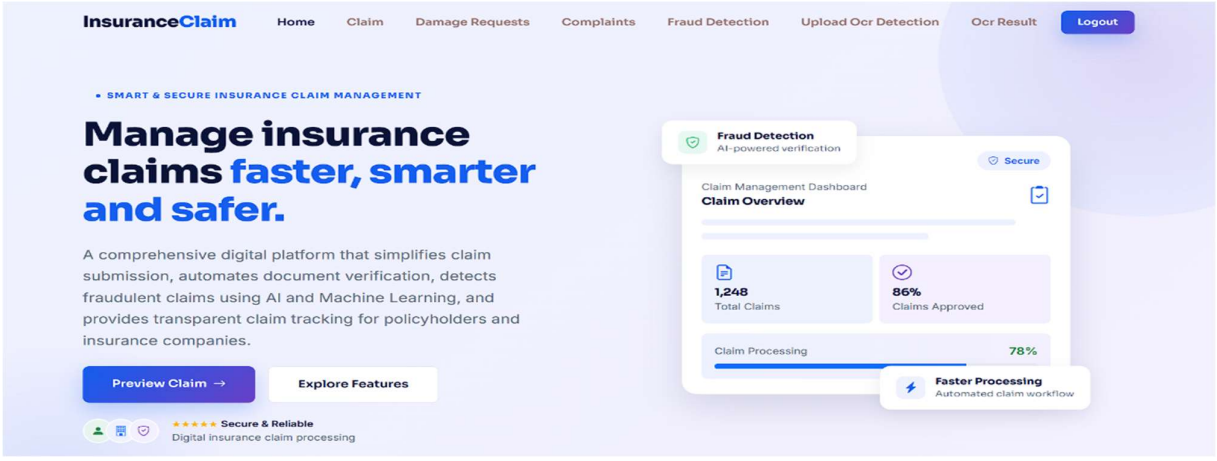


Figure 5.2: Insurance Company Management Interfaces

5.1.3 Claim Management Module

The Claim Management Module handles the complete claim submission process. Users enter claim details and upload supporting documents and vehicle damage images. The system stores the claim with its current status and allows the insurance company to review the submitted information. Users can also track the progress and final decision of their claims.

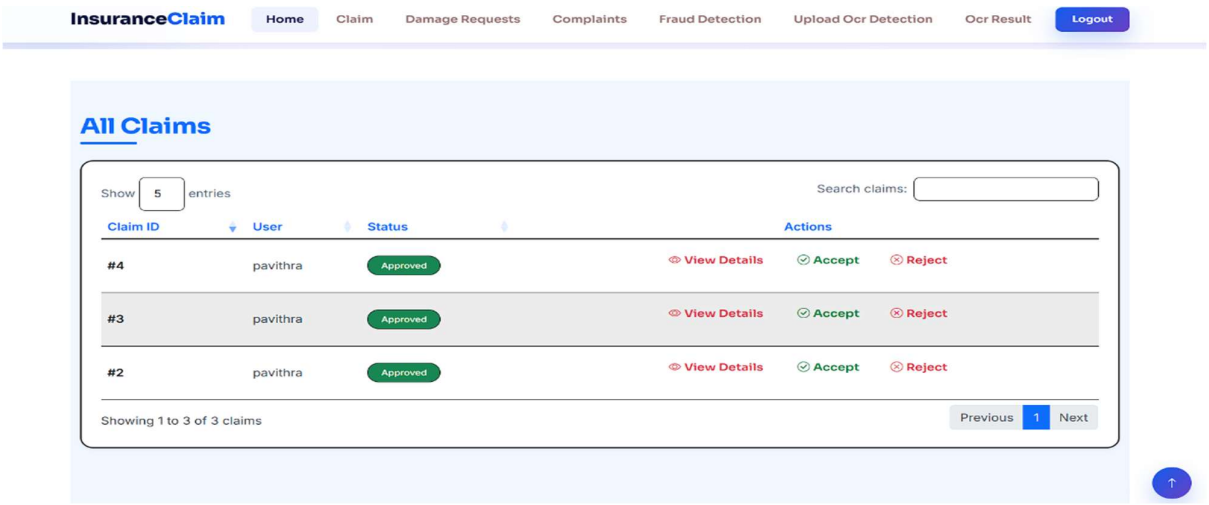
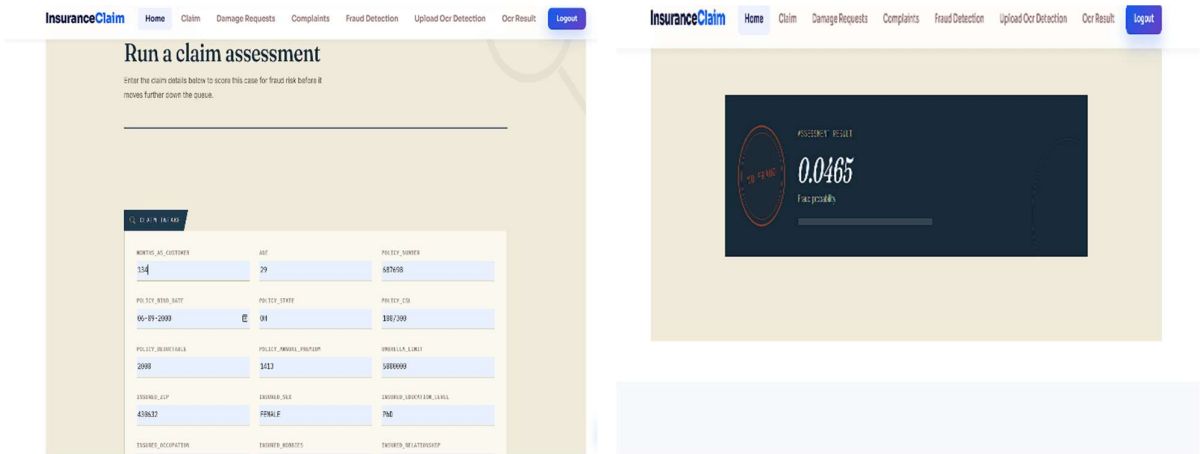


Figure 5.3: Claim Management Interfaces

5.1.4 Fraud Detection and OCR Module

The Fraud Detection and OCR Module processes uploaded insurance documents. The OCR component extracts text and relevant information from PDF, DOCX and image-based documents. The extracted information is processed by the machine-learning fraud detection model, which generates a fraud prediction and probability. The results are stored and made available to the insurance company for verification.



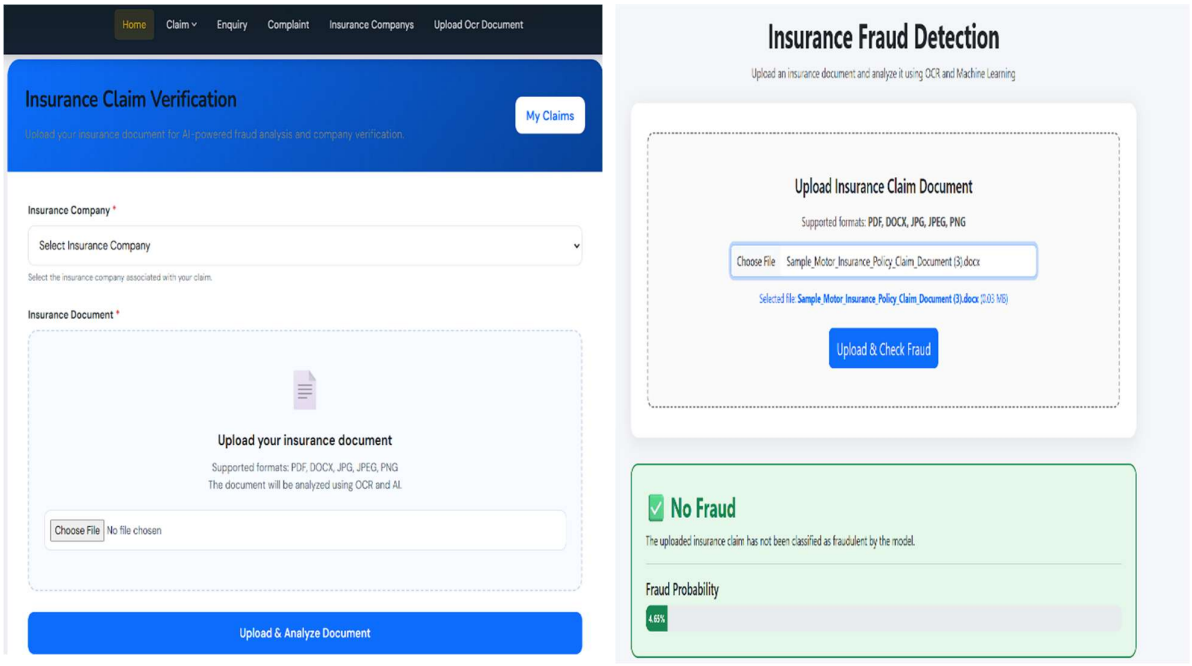
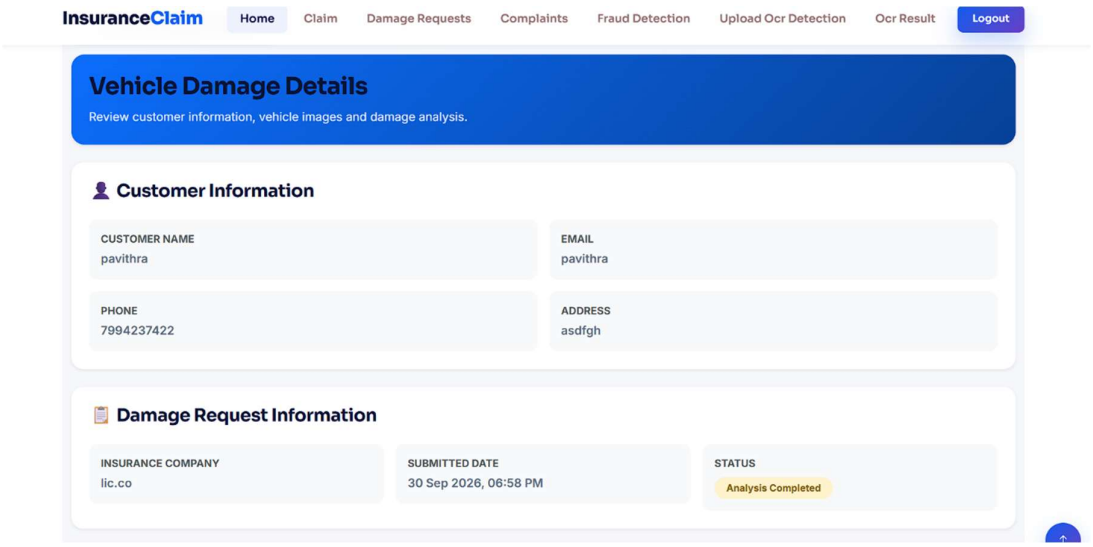


Figure 5.4: Fraud Detection and OCR Interfaces

5.1.5 AI Vehicle Damage Analysis Module

The AI Vehicle Damage Analysis Module analyzes the vehicle images uploaded by the user. Images from different vehicle views such as front, back, left, right, top and bottom are submitted to the AI-based analysis component. The system generates an overall description of the visible vehicle damage, which is stored with the claim and presented to the insurance company during claim review.



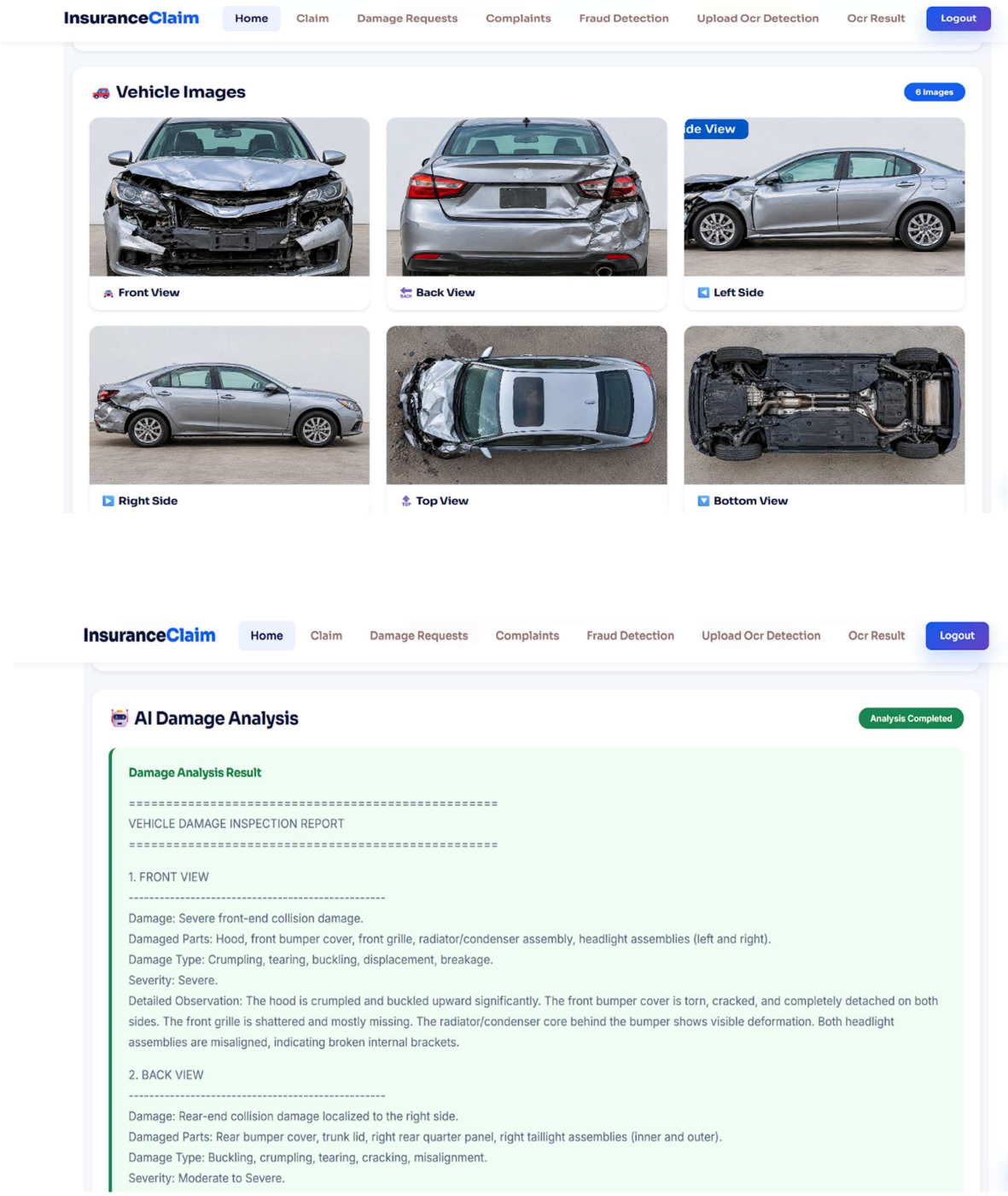


Figure 5.5: AI Vehicle Damage Analysis Interfaces

5.1.6 Admin Management Module

The Admin Management Module provides administrative control over the system. The administrator can view registered users and insurance companies, approve or reject user registrations, monitor complaints and manage system-related information. The module also supports viewing dashboard information and system statistics.

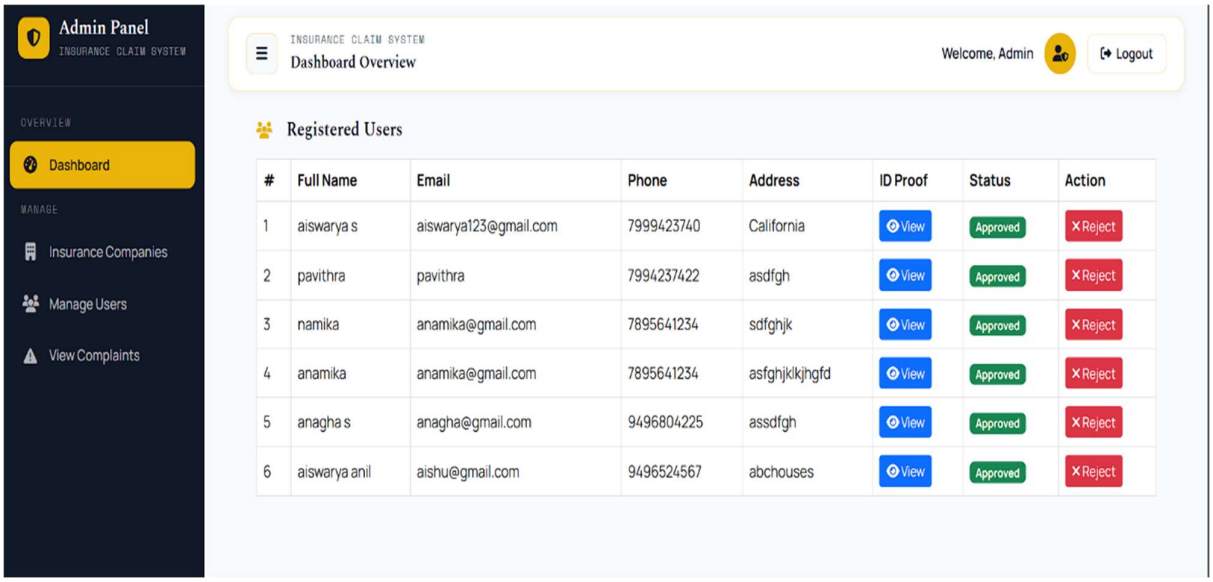


Figure 5.6: Admin Module Interfaces

5.2 Algorithms

The following algorithms describe the main processing procedures of the AI-Powered Smart Insurance Claim Management System. Each algorithm defines the sequence of steps used to perform authentication, claim processing, AI-based analysis, fraud detection, and final claim verification.

5.2.1 User Authentication Algorithm

The process of validating user credentials and checking account approval before granting system access. It ensures that only valid and approved users can access the appropriate dashboard.

Steps:

1. Accept user login credentials.
2. Validate the entered information.
3. Check the credentials against the stored user details.
4. Check the user's approval status.
5. If valid and approved, allow login.
6. Redirect the user to the appropriate dashboard.
7. Otherwise, display an appropriate error message.

5.2.2 Insurance Claim Submission Algorithm

This algorithm describes the process of submitting an insurance claim along with the required documents and vehicle damage images. It validates and stores the claim information before making it available for insurance-company review.

Steps:

1. Accept claim details from the user.
2. Validate the submitted information.
3. Upload supporting documents and damage images.
4. Create the claim record.
5. Store the uploaded files.
6. Set the claim status to Pending.
7. Make the claim available for insurance-company review.

5.2.3 OCR and Fraud Detection Algorithm

This algorithm describes how uploaded insurance documents are processed using OCR and machine learning. The extracted information is analyzed to generate a fraud prediction and probability for further verification.

Steps:

1. Receive the uploaded insurance document.
2. Identify the document type.
3. Extract text using the OCR engine.
4. Extract the required claim-related fields.
5. Prepare the extracted information for the ML model.
6. Apply the trained fraud detection model.
7. Generate the fraud prediction and probability.
8. Store the OCR text and prediction result.
9. Display the result to the insurance company.

5.2.4 AI Vehicle Damage Analysis Algorithm

This algorithm describes the AI-based analysis of uploaded vehicle damage images. The AI component examines the images and generates an overall description of the visible vehicle damage.

Steps:

1. Receive the uploaded vehicle damage images.
2. Validate the six image inputs.
3. Send the images to the AI analysis component.
4. Analyze the visible vehicle damage.
5. Generate an overall damage description.
6. Store the analysis result.
7. Display the result to the insurance company for review.

5.2.5 Claim Verification and Decision Algorithm

This algorithm describes the final claim review and decision-making process. The insurance company uses the claim details, OCR results, fraud prediction, and damage analysis to verify the claim and record the final decision.

Steps:

1. Retrieve the submitted claim.
2. Retrieve OCR and fraud detection results.
3. Retrieve the AI-generated damage analysis.
4. Display the collected information to the insurance company.
5. Review the claim and supporting evidence.
6. Verify or reject the submitted documents.
7. Approve or reject the claim.
8. Store the decision and verification details.
9. Display the final claim status to the user.

These algorithms work together to support the main operations of the AI-Powered Smart Insurance Claim Management System and provide a structured process for submitting, analyzing, verifying, and managing insurance claims.

5.3 Architecture

The architecture of the AI-Powered Smart Insurance Claim Management System is organised into four logical layers: the **Presentation Layer, Application and Processing Layer, AI and ML Intelligence Layer, and Data Storage Layer**, as shown in Figure 5.8. These layers work together to provide a structured and integrated claim-processing workflow. The Policyholder, Insurance Company and System Administrator interact with the system through their respective portals. The Presentation Layer provides user interfaces for claim submission, policy management, claim tracking, document verification and system administration. The Django Application Layer manages authentication, role-based access control, request routing and the core business logic of the system. The AI and ML Intelligence Layer performs vehicle damage analysis, document OCR and fraud prediction, while the Data Storage Layer maintains structured claim information and securely stores images and supporting documents.

The **Policyholder Portal** allows users to register and log in securely, submit insurance claims, provide policy and claim details, upload required documents and vehicle images, track the progress of submitted claims and view available insurance companies. Users can also make enquiries or submit complaints through the system. The portal provides a centralised interface through which policyholders can monitor their claim without depending entirely on manual communication with the insurance company.

The **Insurance Company Portal** provides authorised company personnel with access to claim-related operations. Company users can manage relevant policy information, view submitted claims, examine uploaded documents, review OCR-extracted information and analyse vehicle damage. They can also review fraud-prediction results and verify supporting documents before making an appropriate claim decision. This portal brings the important information required for claim assessment into a single workflow.

The **System Administrator Dashboard** provides overall system management facilities. Administrators can manage users and insurance companies, monitor complaints, supervise system activities and view relevant system statistics. Role-based access control ensures that each category of user can access only the functions associated with their responsibilities, improving both usability and security.

The Application and Processing Layer is implemented using the Django framework. It coordinates URL routing, authentication, role-based access control, form processing, database communication and the core business logic of the system. When a claim is submitted, the application validates the provided information and stores the relevant details before passing the required data to the appropriate

AI and ML components. The application layer therefore acts as the central communication point between the user interfaces, processing modules, intelligence components and database.

The submitted claim information is processed through the **AI and ML Intelligence Layer**. The **Visual Damage Analysis** component analyses vehicle images captured from multiple angles to identify and assess visible damage. The **Document OCR Engine** extracts relevant text and fields from uploaded insurance documents and identity proofs using OCR and document-processing techniques. The **Decision Tree Classifier** uses available claim information and extracted features to perform fraud prediction and generate a corresponding classification or probability result. These intelligent components assist insurance-company personnel by providing automated analysis during the claim assessment process.

The processed information is maintained in the **Data Storage Layer**. A MySQL relational database stores structured information such as user profiles, insurance companies, policies, claims, complaints and analysis-related records. A separate media storage area is used for vehicle damage images, identity proofs, insurance documents and other uploaded files. Maintaining structured records separately from large media files helps organise claim information and supports future expansion of the system.

The overall architecture follows a sequential but interconnected workflow. A policyholder first submits a claim and uploads the required information and supporting files. The Django application processes and stores the submitted data, after which the appropriate OCR, fraud-detection and vehicle-damage analysis components process the available information. The generated results are then made available to authorised insurance-company personnel for verification and decision-making. The final claim status and related information are stored in the database and can be accessed by the relevant users through their respective portals.

The combination of these four architectural layers provides a structured framework for integrating claim management with artificial intelligence and machine learning. It separates user interaction, application processing, intelligent analysis and data management while allowing the components to communicate as part of a single workflow. This architecture provides the foundation for online claim submission, automated document processing, vehicle damage assessment, fraud detection, verification and claim tracking within the proposed system.

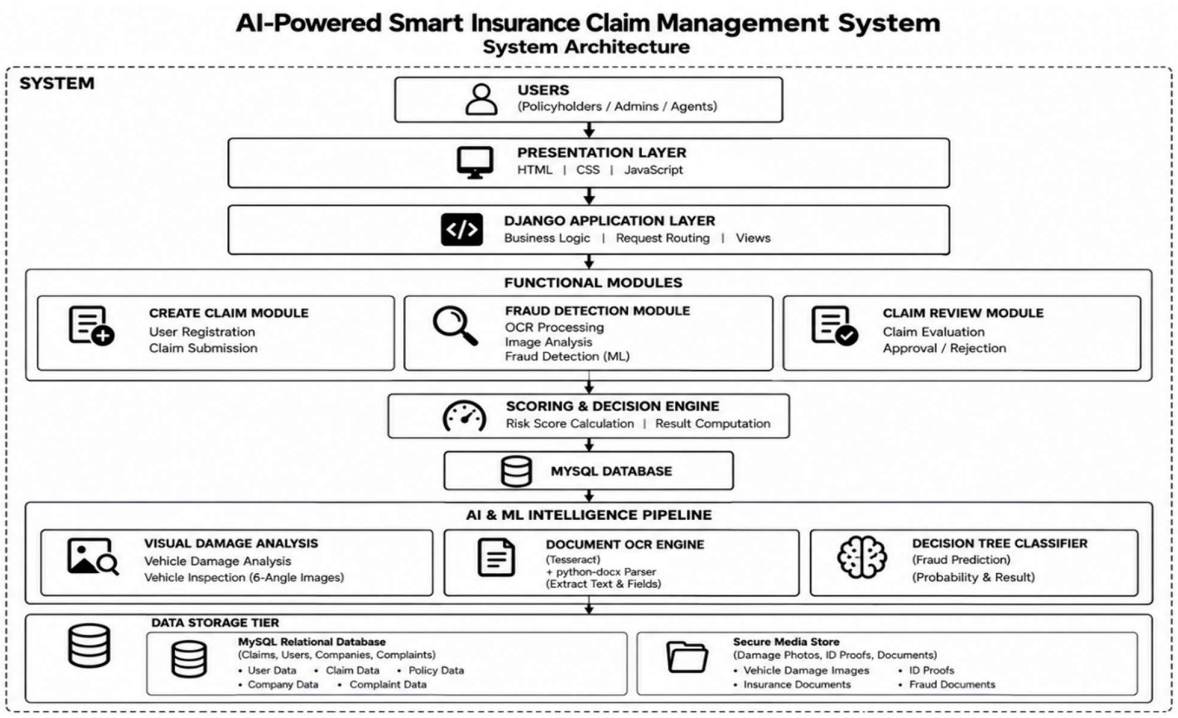


Figure 5.7: System Architecture of the AI-Powered Smart Insurance Claim Management System

5.4 Case Study

The AI-Powered Smart Insurance Claim Management System can be illustrated through a typical motor insurance claim submitted by a policyholder after a vehicle accident. Consider a situation where a policyholder's vehicle is involved in a road accident and the owner needs to submit an insurance claim. Instead of depending entirely on manual claim processing, the policyholder accesses the system through the Policyholder Portal and logs into the platform.

The policyholder provides the required claim information and uploads supporting documents such as the insurance policy, identification proof and other relevant documents. The user also uploads photographs of the damaged vehicle from multiple angles. The submitted information and files are securely transferred to the application layer, where the Django-based system validates the request and stores the relevant information.

Once the claim is submitted, the Document OCR Engine processes the uploaded documents. OCR and document-parsing techniques are used to extract important information such as policy details, personal information and other relevant fields. This reduces the need for manual entry and allows the extracted information to be used during subsequent claim processing.

At the same time, the uploaded vehicle photographs are processed by the Visual Damage Analysis component. Images captured from different angles are analysed to identify visible vehicle damage and provide an automated assessment that can assist the insurance company during claim evaluation.

The analysis can help identify damaged areas and provide supporting information for the assessment of the submitted claim.

The system also applies the Decision Tree Classifier as part of the fraud detection process. Relevant claim information, extracted document details and other available features are analysed to identify patterns associated with potentially fraudulent claims. The classifier produces a fraud prediction or probability result that can be considered by authorised insurance personnel during the claim review process. The automated result is intended to support the review process rather than replace the final decision of the insurance company.

The Insurance Company Portal allows authorised personnel to view the submitted claim, verify the uploaded documents, examine the vehicle damage analysis and review the fraud detection result. The company can then evaluate the complete claim information and proceed with the appropriate claim-processing decision.

The System Administrator Dashboard provides administrative monitoring facilities, including user management, complaint monitoring and system statistics. The administrator can oversee the operation of the platform while maintaining the separation of responsibilities between policyholders, insurance companies and administrators.

After the claim has been reviewed, the relevant claim information and processing results are maintained in the MySQL database, while the uploaded vehicle images and supporting documents remain in the secure media store. The policyholder can subsequently access the portal to track the status of the claim.

This case study demonstrates how the AI-Powered Smart Insurance Claim Management System supports the complete insurance claim workflow within a single platform, from claim submission and document processing to vehicle damage analysis, fraud detection, claim evaluation and status tracking. By integrating conventional web-based claim management with AI-assisted analysis, OCR and machine-learning-based fraud prediction, the system provides a structured approach to improving the efficiency and automation of insurance claim processing.

CHAPTER 6
IMPLEMENTATION DETAILS

6.1 Implementation Procedures

The implementation of the AI-Powered Smart Insurance Claim Management System with Fraud Detection and Automation was carried out in a series of stages so that the different components could be developed and integrated in an organised manner. The process began with analysing the requirements of an insurance claim management system and identifying the functions needed by users, insurance companies and administrators. Based on these requirements, the system architecture, user roles, database structure and major modules were planned before development. The system was implemented as a web-based application using Python and the Django framework for backend development. Django handles application logic, request handling, URL routing, database operations and views. HTML, CSS and JavaScript were used to build the user-friendly frontend, with separate interfaces for users, insurance companies and the administrator. MySQL was used as the database to store and retrieve application data. The required libraries for machine learning, OCR and image analysis were configured along with the web framework.

After the environment was configured, the database models were created. The User profile model stores user registration details and approval status, and the Insurance company model stores company details along with licence proof. The Claim model is the central model, with about 45 fields covering policy, claimant, incident, vehicle and claim amount details. The Complaint model stores user complaints and administrator replies. The Damage_image_check model stores the six vehicle photographs submitted with a claim. The UserFraudDocument model stores the uploaded document, OCR text, extracted fields, the ML prediction and the company's verification result.

The authentication and role-based access functions were implemented next for the three roles: administrator, insurance company and user. Users register and wait for administrator approval before they can log in, and insurance companies register with licence proof, which the administrator reviews before approval. Passwords are stored using Django's password hashing, and each role is directed to its own dashboard after login.

The claim submission module was then integrated with document upload. An approved user submits a claim by entering the policy, incident, claim amount and vehicle details and uploading the supporting files. The information is stored in the database and the claim is initially assigned a Pending status, so the insurance company can review it before making a decision. Users can track the progress of their claims from the dashboard.

The Optical Character Recognition (OCR) module (fraud_ocr.py) was incorporated to extract relevant information from uploaded insurance documents, reducing the need to enter all document data manually. The system extracts text from the document and then identifies key fields from that text. The OCR output, extracted fields and processing status are stored in the UserFraudDocument model for verification and fraud analysis.

The machine-learning-based fraud detection module identifies potentially suspicious claims and documents. The model was trained on a dataset of 1000 claim records with 38 input features and the target attribute fraud_reported (247 fraudulent and 753 non-fraudulent cases). It is a scikit-learn Pipeline in which a ColumnTransformer imputes missing values (median for numeric and most frequent for categorical features) and applies one-hot encoding, followed by a DecisionTreeClassifier (random_state 42). The trained pipeline is saved with joblib as fraud_detection_model.pkl and loaded by the Django views. When claim details are submitted, they are converted into the model's input format, and the prediction, along with its probability and processing status, is stored for further examination. The output is intended to support the verification process, not to independently establish that a claim is fraudulent.

The damage image analysis module (analyze_damage.py) uses the API to process the six vehicle photographs attached to a claim and produce a description of the visible damage. This gives the insurance company additional information for checking whether the reported damage is consistent with the claim.

The company verification module lets insurance company staff log in and view submitted claims together with the ML prediction, OCR results and damage analysis. They can verify or reject each claim, and the decision updates the claim status, which the user can see. The system can also generate a downloadable analysis report summarising the claim details and fraud analysis results.

The administrator module manages users, companies, claim records and other system functions. The complaint management facility lets users raise complaints, and the administrator can review and reply to them.

Finally, the modules were integrated into a unified claim management workflow that combines online claim submission, document processing, fraud detection, damage image analysis, claim verification and status tracking in one application. Each module was developed and tested independently before being connected to the others. The completed implementation provides a digital platform that simplifies insurance claim management, reduces manual processing and supports more efficient claim verification.

6.2 Testing Methods and Results

Testing was carried out throughout development to verify that the different modules function correctly and work together as expected. The tests used sample data representing the activities normally performed during insurance claim processing, with both valid and invalid inputs to check the system under different conditions.

6.2.1 Testing Dataset

The testing dataset was prepared specifically for this application and consisted of sample records for the different modules:

- User data: Admin, Insurance Company and User accounts, with registration details, company licence proofs and approval requests.
- Claim data: sample claim records with policy details, incident details, vehicle details and claim amounts.
- Document data: sample claim documents, including clear scans, blurred images, and documents with missing or mismatched details, for testing OCR text extraction.
- Damage image data: sets of six vehicle photos showing different levels of damage.
- Fraud prediction data: claim records with genuine and suspicious values taken from the fraud detection dataset.

This dataset was used only for testing and does not represent real customer information.

6.2.2 Testing Methods

The main testing strategies were unit testing, integration testing, functional testing and system testing.

- Unit testing verified individual functions such as registration, login, claim submission, document upload and fraud prediction.
- Integration testing checked whether modules exchange data correctly, for example whether a submitted claim is passed to the fraud prediction model and the result is then displayed to the insurance company.
- Functional testing verified that each feature produces the expected result for given inputs.
- System testing evaluated the complete workflow from user registration to final claim verification.

Validation testing was also performed to confirm that incorrect and incomplete inputs are rejected with appropriate messages.

6.2.3 Test Results

Sl. No.	Module	Test Case	Expected Result	Result
1	Registration	User registers with valid details	Account created and set to pending approval	Pass
2	Login	Unapproved user attempts login	Login denied until the admin approves	Pass
3	Registration	Company registers with licence proof	Company stored and awaits admin approval	Pass
4	Claim	Submit claim with all required fields and images	Claim stored with all fields and six images, initial status set	Pass
5	Validation	Submit claim with missing or invalid fields	Submission rejected with error message	Pass
6	Fraud ML	Predict on a valid claim record	Model loads and returns a fraud / non-fraud prediction	Pass
7	OCR	Upload a document	Text extracted and key fields saved	Pass
8	Damage analysis	Upload vehicle images	Damage description generated and stored	Pass
9	Company	Company verifies or rejects a claim	Claim status updated and visible to the user	Pass

Table 6.1: Test Cases and Results

The claim module was tested with the prepared claim dataset to verify successful submission, validation of required fields and correct storage of claim details. The fraud prediction feature was tested with different claim records to check whether the trained model returned a fraud or non-fraud prediction for each input. The OCR module was tested with different document images to verify that text was extracted and that fields such as names, dates and policy details were correctly identified. The damage analysis feature was tested with different vehicle photos to check that useful observations about the damage were generated. The role-based features were tested using separate Admin, Insurance Company and User sessions.

The testing therefore verified registration and approval, claim submission, input validation, fraud prediction, document OCR, damage image analysis and company verification. The testing activities verified that:

- Registration, approval and login validation work as intended.
- Claims are stored with complete details and supporting images.
- Incorrect or incomplete inputs are rejected.
- The ML model produces a fraud prediction for submitted claims.
- OCR extracts and stores text and key document fields.
- Damage images are analysed and the result is stored.
- Companies can verify or reject claims and the status is updated.

The results showed that the major functions of the system worked according to the expected requirements. Issues identified during testing were corrected and the affected modules were tested again to confirm proper operation.

CHAPTER 7
ADVANTAGES AND DISADVANTAGES

7.1 Advantages

The Insurance Claim Management and Fraud Detection System provides several advantages by combining online claim submission, document verification, damage image analysis, machine-learning-based fraud prediction, and administrative management into a single platform. The system helps organize the claim handling process and reduces the need for separate manual verification procedures. By integrating claim details with OCR-based document processing and fraud prediction, the system provides a structured approach for evaluating insurance claims. The major advantages of the proposed system are listed below.

- Provides a single platform for managing users, insurance companies, claims, complaints, documents, and fraud analysis.
- Allows users to register, submit insurance claims online, and track the status of their claims without visiting the insurance office.
- Provides role-based access for Admin, Insurance Company, and User, so each user can access only the functions meant for their role.
- Ensures that users and insurance companies are verified by the administrator before accessing the system, with companies required to submit licence proof during registration.
- Predicts the possibility of fraud in a claim using a trained machine learning model that analyses the claim details.
- Performs automated document processing by extracting text and relevant fields from uploaded claim documents using OCR.
- Analyses the uploaded vehicle damage photos using AI-based image analysis, helping insurance companies assess the reported damage.
- Reduces the manual effort involved in checking claim documents, comparing details, and identifying suspicious claims.
- Allows insurance companies to review the fraud prediction, extracted document details, and damage analysis before verifying or rejecting a claim.
- Maintains the claim status throughout the process so that users can view the current stage of their claims.
- Provides a complaint management feature through which users can register complaints and receive replies from the administrator.

- Generates a downloadable analysis report for each claim, which helps in record keeping and decision making.
- Maintains user, claim, document, and analysis information in a centralized database, making the information easier to retrieve and manage.

7.2 Disadvantages

Although the Insurance Claim Management and Fraud Detection System provides several useful features for claim handling and fraud detection, the current implementation has certain limitations. These limitations are mainly related to the accuracy of OCR and image analysis, dependence on the quality of the data provided by users, and the scope of the current fraud detection model. Since fraud prediction is based on a model trained on a limited dataset, the results may not completely represent every type of fraudulent claim. The current version also focuses mainly on vehicle insurance claims and can be further enhanced in future versions. The major disadvantages of the current system are listed below.

- The accuracy of OCR-based document processing depends on the quality, clarity, and format of the uploaded documents.
- The fraud prediction result depends on the dataset used for training, which contains a limited number of claim records with fewer fraudulent cases than genuine ones, so some fraudulent claims may not be detected.
- The machine learning model gives a prediction only and does not completely replace human decision-making, since the final verification or rejection is performed by the insurance company.
- The damage image analysis depends on the clarity, angle, and lighting of the uploaded photos and may not always give an accurate assessment.
- The damage analysis feature depends on an external AI service and an internet connection, so it may not work when the service is unavailable.
- The current system is mainly designed for vehicle insurance claims and may require additional development to support other insurance types such as health or life insurance.
- The system relies on the information entered by users and companies, so incorrect or incomplete data can affect the quality of the analysis.
- The system does not include features such as real-time integration with police, hospital, or government databases for verifying claim details.

- The current implementation is designed for the present claim workflow and may require additional development and a production-grade database to support a very large number of concurrent users.

CHAPTER 8
RESULT AND DISCUSSION

8.1 Analysis

The analysis of the AI-Powered Smart Insurance Claim Management System with Fraud Detection and Automation was carried out to evaluate the overall functionality and effectiveness of the web-based claim management platform. The evaluation examined how the different modules operate individually and how they work together as a complete system for claim submission, automated fraud analysis and company verification. The analysis mainly considered user and company registration, administrative approval, claim submission, document processing, fraud prediction, damage image analysis, company verification and claim status management.

One of the important aspects analysed was the registration and approval process. The system allows users and insurance companies to register, and access is granted only after the administrator approves the account. Companies submit licence proof during registration, which the administrator can check before approval. This provides a controlled method of managing access and helps ensure that only verified users and companies take part in the claim process.

The claim submission module was analysed to verify how claim information is collected and stored. Users enter policy, incident, vehicle and claim amount details and upload supporting files and vehicle images. The claim record, which holds about 45 fields, is stored with an initial Pending status. Users can track the progress of their claims, and companies can review each claim before making a decision. This replaces manual paperwork with a structured digital process.

The fraud detection module was analysed to examine how claim data is processed by the machine learning model. The model is a decision tree classifier trained on a dataset of 1000 claim records with 38 features, of which 247 were fraudulent and 753 non-fraudulent. Missing values are handled and categorical features are encoded within the same pipeline, so submitted claim details can be passed to the model directly. The system returns a fraud or non-fraud prediction for each claim and stores it with the claim record. The analysis confirmed that the prediction acts as a support for the insurance company's review and is not a final decision on the claim.

The OCR document analysis module was evaluated to examine how information is extracted from uploaded documents. The system extracts text from the document and then identifies key fields, which are stored with the document record. This reduces the effort of reading documents manually and gives the company structured information for verification and fraud analysis.

The damage image analysis module was also analysed. The six vehicle photographs submitted with a claim are processed using the API, and a description of the visible damage is generated. This gives

the company additional information for checking whether the reported damage is consistent with the claim.

The company verification module was analysed to examine the claim review process. Insurance company staff can view submitted claims together with the ML prediction, OCR results and damage analysis, and can verify or reject each claim. The claim status is updated according to the decision, so users can see the outcome of their claim. A downloadable analysis report is also available, summarising the claim details and the fraud analysis results.

The administrative module and supporting features were analysed as well. The administrator manages users, companies and claim records, and can view and reply to complaints raised by users. These features support the main claim workflow and improve communication between users and the platform.

The overall usability of the system was also analysed. The system provides separate interfaces for the administrator, insurance companies and users, each with the functions relevant to its role. This separation organises the claim workflow and keeps each role to its own functions. The modular structure also improves maintainability, because user management, claim handling, fraud prediction, OCR, image analysis and company verification are implemented as separate components that communicate through the application and the database.

The key observations obtained from the analysis include:

- The system supports registration of users and companies with administrator approval.
- Companies can register with licence proof for verification.
- Users can submit claims with detailed information and supporting vehicle images.
- Claim records are stored with an initial status and can be tracked by users.
- The ML model generates a fraud or non-fraud prediction for submitted claims.
- Uploaded documents can be processed by OCR to extract text and key fields.
- Vehicle damage images can be analysed to produce a description of visible damage.
- Insurance companies can review claims along with the fraud analysis and verify or reject them.
- The claim status is updated according to the company's decision.
- Users can raise complaints and receive administrator replies.
- A downloadable analysis report summarises the claim and fraud analysis details.
- The modular structure allows the different components to operate as a unified system.

Overall, the analysis indicates that the system successfully integrates user management, claim submission, document processing, machine learning based fraud detection, damage image analysis

and company verification into a single web-based platform. It provides a structured workflow for processing a claim from submission through automated analysis to the final decision.

The results demonstrate that combining several analysis components in one platform can organise the claim process and reduce the need to review documents, images and claim data through separate procedures. By combining the ML fraud prediction, OCR extraction and damage image analysis with a company verification interface, the system supports a more systematic and efficient approach to claim verification, while leaving the final decision with the insurance company.

8.2 Integrated Discussion

The combined results show that the main strength of the proposed system is the integration of several supporting technologies into one claim-management workflow. Instead of treating claim submission, document reading, fraud screening and vehicle-damage assessment as separate activities, the system connects them through a common claim record. This gives the insurance company a single place from which to review the evidence associated with a claim.

The machine-learning component should be interpreted as a screening mechanism rather than an autonomous claim-approval mechanism. The Decision Tree classifier produces a fraud or non-fraud prediction and a probability that can be considered during review. This human-in-the-loop approach is important because automated predictions can be affected by the quality and distribution of the training data. The final decision remains with authorised insurance-company personnel.

The OCR component contributes by converting information contained in uploaded documents into text and extracted fields that can be stored and reviewed. Its usefulness depends on document quality, layout and image clarity. When OCR output is combined with the original document and the entered claim details, the reviewer has more structured information for checking consistency instead of relying only on manual reading.

The vehicle damage analysis component provides another form of decision support. The six submitted vehicle views are analysed to generate a description of visible damage. This does not replace physical inspection or professional assessment; rather, it provides an additional source of information that can be compared with the claim description and supporting evidence.

From a system-design perspective, the role separation also contributes to the reliability of the workflow. Users submit and track claims, insurance companies review evidence and make decisions, and administrators manage approvals and complaints. This separation reduces unnecessary overlap between responsibilities and provides a clearer audit trail of the actions performed during claim processing.

The implementation therefore meets the central project objective of demonstrating an integrated AI-assisted insurance claim platform. The current evaluation is functional rather than a large-scale

production benchmark, so the results should not be interpreted as proof of a specific real-world fraud-detection accuracy. Future evaluation with larger, representative datasets and measured OCR and image-analysis performance would provide stronger evidence for deployment.

8.3 Objective-wise Evaluation

The first objective of the project is to provide a centralised platform for insurance claim submission and tracking. The implemented workflow allows an approved user to enter claim information, upload supporting documents and vehicle photographs, and monitor the claim status. The insurance company can access the same claim record for review and decision-making. This addresses the fragmentation found in conventional paper-based and email-based processing.

The second objective is to support automated document processing and fraud screening. The OCR module extracts text and relevant fields from uploaded documents, while the machine-learning pipeline uses structured claim information to generate a fraud or non-fraud prediction. The result is stored with the claim so that authorised insurance personnel can examine it together with the original evidence. The prediction is therefore used as decision support rather than as an automatic final decision.

The third objective is to support vehicle-damage assessment through image analysis. Six vehicle views are accepted and processed to generate a description of visible damage. The output provides additional information during company review and can help the reviewer compare the reported incident with the visible evidence. It does not replace professional inspection or the final judgement of the insurance company.

The final objective is to provide administrative control and a structured role-based workflow. The administrator can approve users and companies, monitor complaints and review system information. Separating the responsibilities of users, insurance companies and administrators provides a clearer sequence from registration to claim decision and supports more organised management of the overall claim process.

8.4 Security and Data Handling

The system handles user details, vehicle information, claim descriptions, identity documents, insurance documents and damage photographs. These records require controlled access because different participants should not be able to view or modify information outside their responsibilities. The application therefore uses role-based access and separates the operations available to the User, Insurance Company and Administrator.

Authentication and approval form an important part of the access-control workflow. Users and insurance companies register before using the system, and the administrator can approve their accounts. Django authentication and password handling are used for login, while the role associated with an account determines the functions and dashboard available after authentication.

Claim documents and images are associated with the relevant claim so that the original evidence can be reviewed together with extracted OCR information, fraud-prediction results and vehicle-damage analysis. The company verification process records the verification result and claim decision. This improves traceability compared with informal processing in which evidence and decisions may be distributed across emails, paper records and separate files.

The present prototype should not be interpreted as a complete production-security implementation. A deployed system would require encrypted transport, secure file validation, access logging, backup and recovery procedures, stronger session controls, vulnerability assessment and regular security testing. These measures form an important part of future deployment and would help protect the confidentiality and integrity of insurance claim information.

8.5 Usability, Maintainability and Scalability

The role-based organisation of the system improves usability by providing different functions according to the responsibilities of users, insurance-company personnel and administrators. Users can register, submit insurance claims, upload supporting documents and vehicle photographs, track claim status and raise complaints. Insurance-company personnel can review claims, examine OCR-extracted information, check fraud-prediction results, view vehicle-damage analysis and record claim decisions. Administrators can manage registrations, monitor system activities and handle complaints. This separation makes the system easier to understand and prevents users from accessing unnecessary functions.

The modular structure of the application supports maintainability by separating major functions such as user management, claim management, document upload, OCR processing, fraud prediction, vehicle-damage analysis and administration. This makes the system easier to test, debug and modify because changes to one component can be made with minimal effect on other modules. The machine-learning components can also be updated or retrained independently when improved data or models become available. Structured database relationships further support maintenance by organising users, insurance companies, claims, documents and analysis results efficiently.

Scalability is an important consideration for a production-level insurance environment. The current prototype uses Django and MySQL and stores uploaded documents and vehicle images along with their associated claim information. For larger deployments, database indexing and query optimisation can improve response time as the amount of data increases. Dedicated or cloud-based storage can also be used to manage the growing number of documents and images while providing suitable backup and recovery options.

Asynchronous processing could further improve scalability. OCR extraction, fraud prediction and vehicle-damage analysis may require more processing time than normal web requests. These operations could therefore be handled as background tasks, allowing users and insurance-company personnel to continue using the system while analysis is being performed. The system could display statuses such as Pending, Processing and Completed to indicate the progress of each operation.

The present project mainly focuses on functional integration rather than large-scale performance evaluation. Therefore, performance under thousands of concurrent users has not been experimentally established. Future testing should measure claim-submission response time, database response time, OCR processing time, image-analysis time, fraud-prediction time, file-upload performance and system recovery. Load and stress testing can also help identify performance limitations as the number of users and claims increases.

Overall, the role-based design improves usability, while the modular architecture supports maintainability and future development. The current Django and MySQL implementation provides the basic foundation, while database optimisation, improved storage, background processing and scalable infrastructure can support future expansion into a larger insurance claim-management platform.

CHAPTER 9
CONCLUSION

9.1 Summary

The AI-Powered Smart Insurance Claim Management System with Fraud Detection and Automation was developed as a web-based platform designed to organise and streamline the insurance claim process. The system integrates several important functions, including user and insurance company registration, administrative approval, claim submission, document processing using OCR, machine learning based fraud prediction, vehicle damage image analysis, company-level claim verification, complaint management and claim status tracking. These features work together to provide a structured workflow from claim submission to the final decision by the insurance company.

The development of the system demonstrates how web technologies and artificial intelligence can be used to support insurance claim processing and reduce repetitive manual work. By using Django for application development and combining it with a machine learning model, OCR and image analysis, the system provides a centralised platform for collecting and examining claim information. The fraud detection component analyses the submitted claim details and predicts whether a claim is likely to be fraudulent, the OCR module extracts key information from uploaded documents, and the damage analysis module describes the damage visible in the submitted vehicle photographs.

Another important aspect of the system is the integration of these analysis results for the insurance company. Company staff can view the claim details, the fraud prediction, the extracted document fields and the damage analysis in one place. Based on this information, they can verify or reject the claim, after which the system updates the claim status so that the user can see the outcome. A downloadable analysis report is also available, summarising the claim and the fraud analysis.

The results obtained during testing and analysis indicate that the implemented modules support the intended claim workflow. The system successfully provides registration and approval, claim submission, fraud prediction, document OCR, damage image analysis, company verification and claim status tracking. The modular design also provides a structured foundation for extending the system with additional features in the future.

Overall, the system demonstrates how an integrated web-based platform can support a systematic approach to insurance claim management and fraud screening, while leaving the final decision with the insurance company.

9.2 Limitations

The system covers almost the complete insurance claim lifecycle in a single platform, including registration and approval, claim submission, OCR document analysis, machine learning based fraud prediction, damage image analysis, company verification, complaint management, claim status tracking and downloadable reports. The limitations identified are therefore refinements of existing features and do not point to missing core functionality. The fraud prediction model was trained on 1000 claim records, which is enough to demonstrate the complete fraud screening workflow. Its accuracy can be improved further by retraining on larger, real-world data, and since the model is saved as a pipeline, it can be replaced without changing the rest of the system. The OCR and damage analysis modules give their best results with clear documents and photographs, and their output serves as supporting information that the company checks during verification. The final decision to verify or reject a claim is made by the insurance company, with the automated analysis as decision support. This is a deliberate design choice, as it combines automated analysis with human judgement. The system currently focuses on vehicle insurance claims, and its modular structure allows other insurance types to be added in the future. Overall, the limitations are minor compared with the breadth of functionality delivered, and they show where the system can be improved in future versions.

9.3 Future Scope

The system has several opportunities for further improvement and expansion. As artificial intelligence and insurance technologies continue to develop, additional features can be integrated to improve the accuracy and capabilities of the system.

One possible improvement is to train the fraud detection model on a larger and more diverse dataset, ideally with real claim records. More advanced algorithms, such as random forest or gradient boosting, could be evaluated and compared to improve prediction accuracy. The system could also show the main factors behind each prediction, so that company staff can understand why a claim was flagged.

Another potential enhancement is improving the document analysis. More accurate OCR and natural language processing methods could extract information from a wider range of document types and layouts, and could cross-check extracted fields against the details entered in the claim form to detect mismatches automatically.

The damage analysis module could be extended to estimate repair costs and compare the estimate with the claimed amount. The system could also detect reused or manipulated images, which is a common form of claim fraud.

The system could also be extended to other types of insurance, such as health, life and property insurance, with separate claim forms and fraud models. Additional features such as email and SMS notifications, online premium and settlement payment, and a mobile application would further improve the user experience. The system could also use a more scalable database and cloud deployment to support a larger number of users and companies, and provide more detailed reports and analytics for administrators and companies.

Overall, future development can extend the system from a claim submission and fraud screening platform into a more comprehensive insurance management system with advanced fraud detection, automated claim assessment and intelligent customer support.

9.4 Applications

The system can be applied in various insurance environments where organisations need to process claims and detect possible fraud systematically. It is particularly useful for companies that receive a large number of claims and need a structured approach to the initial verification of each claim.

One of the main applications is in insurance companies, especially for vehicle insurance. Companies can use the platform to receive claims online, review the supporting documents and photographs, and use the fraud prediction and damage analysis to support their verification process through a centralised system.

The system can also be useful for small and medium-sized insurance providers that may need a simple claim management platform without developing separate systems for document processing, fraud screening and claim tracking. The integrated workflow allows these activities to be handled in a single application.

Another application is in insurance agencies, brokers and third-party administrators who handle claims on behalf of several insurers. The ability to store claim details, documents and analysis results in one place helps organise their workflow. The system can also support fraud investigation and audit teams, who can use the prediction results, extracted document fields and analysis reports to identify claims that need closer examination.

Policyholders also benefit from the system, since they can submit claims online, track their status, raise complaints without visiting an office.

Overall, the system can be applied in different insurance scenarios where claim information, documents, images, fraud analysis and claim decisions need to be managed in an organised manner. By integrating these activities into a single web-based platform, it provides a practical approach to supporting structured claim processing and fraud screening.

CHAPTER 10

APPENDICES

```
<!DOCTYPE html>
<html lang="en">
<head>
   <meta charset="UTF-8">
   <meta name="viewport" content="width=device-width, initial-scale=1.0">
   <title>Admin Dashboard - Insurance Management System</title>
   <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/css/bootstrap.min.css"
rel="stylesheet">
   <link href="https://cdnjs.cloudflare.com/ajax/libs/font-awesome/6.5.0/css/all.min.css"
rel="stylesheet">
   <link
href="https://fonts.googleapis.com/css2?family=Spectral:wght@500;600;700&family=Manrope:wg
ht@400;500;600;700;800&family=Space+Mono:wght@400;700&display=swap" rel="stylesheet">
   <script src="https://cdn.jsdelivr.net/npm/chart.js@4.4.1/dist/chart.umd.min.js"></script>
   <style>
   </style>
</head>
<body>
   <nav class="sidebar" id="sidebar">
      <div class="sidebar-header">
         <div class="badge-icon"><i class="fas fa-shield-alt"></i></div>
         <div class="header-text">
            <h3>Admin Panel</h3>
            <p>Insurance Claim System</p>
         </div>
      </div>
      <div class="nav-menu">
         <div class="nav-eyebrow">Overview</div>
         <div class="nav-item">
            <a href="/admin_home/" class="nav-link active">
               <i class="fas fa-tachometer-alt"></i>
               <span>Dashboard</span>
            </a>
         </div>
         <div class="nav-eyebrow">Manage</div>
         <div class="nav-item">
            <a href="/admin_view_insurancecompany/" class="nav-link">
               <i class="fas fa-building"></i>
               <span>Insurance Companies</span>
            </a>
         </div>
         <div class="nav-item">
            <a href="/admin_view_users/" class="nav-link">
               <i class="fas fa-users"></i>
               <span>Manage Users</span>
            </a>
         </div>
         <div class="nav-item">
```

```
            <a href="/admin_view_complaint/" class="nav-link">
              <i class="fas fa-exclamation-triangle"></i>
              <span>View Complaints</span>
            </a>
          </div>
        </div>
        <div class="sidebar-footer">
          <p>SAFEGUARD ADMIN &middot; V2.1</p>
        </div>
      </nav>
      <div class="main-content" id="mainContent">
        <div class="top-bar">
          <div style="display:flex; align-items:center; gap:16px;">
            <button class="toggle-btn" title="Toggle Sidebar">
              <i class="fas fa-bars"></i>
            </button>
            <div class="page-heading">
              <span>Insurance Claim System</span>
              Dashboard Overview
            </div>
          </div>
          <div class="admin-info">
            <span class="welcome-text">Welcome, Admin</span>
            <div class="admin-avatar">
              <i class="fas fa-user-shield"></i>
            </div>
            <a href="/login/" class="logout-btn">
              <i class="fas fa-sign-out-alt"></i> Logout
            </a>
          </div>
        </div>
        <div class="stats-section">
          <div class="stats-grid">
            <div class="stat-card">
              <div class="stat-header">
                <div>
                  <div class="stat-number">{{ company_count|default:0 }}</div>
                  <div class="stat-label">Insurance Companies</div>
                </div>
                <div class="stat-icon"><i class="fas fa-building"></i></div>
              </div>
            </div>
            <div class="stat-card">
              <div class="stat-header">
                <div>
                  <div class="stat-number">{{ users_count|default:0 }}</div>
                  <div class="stat-label">Registered Users</div>
                </div>
                <div class="stat-icon"><i class="fas fa-users"></i></div>
              </div>
            </div>
```

```
            <div class="stat-card">
              <div class="stat-header">
                <div>
                  <div class="stat-number">{{ complaint_count|default:0 }}</div>
                  <div class="stat-label">Total Complaints</div>
                </div>
                <div class="stat-icon"><i class="fas fa-exclamation-triangle"></i></div>
              </div>
            </div>
            <div class="stat-card">
              <div class="stat-header">
                <div>
                  <div class="stat-number">{{ reply_count|default:0 }}</div>
                  <div class="stat-label">Pending Replies</div>
                </div>
                <div class="stat-icon"><i class="fas fa-reply"></i></div>
              </div>
            </div>
          </div>
        </div>
        <h2 class="section-title">
          <i class="fas fa-chart-pie"></i> Analytics Overview
        </h2>
        <div class="charts-grid">
          <div class="chart-card">
            <h5>System Overview (Bar)</h5>
            <div class="chart-container">
              <canvas id="barChart"></canvas>
            </div>
          </div>
          <div class="chart-card">
            <h5>Complaint Status (Pie)</h5>
            <div class="chart-container">
              <canvas id="pieChart"></canvas>
            </div>
          </div>
          <div class="chart-card">
            <h5>Users vs Companies (Doughnut)</h5>
            <div class="chart-container">
              <canvas id="doughnutChart"></canvas>  </div>
          </div>
          <div class="chart-card">
            <h5>Monthly Activity Trend (Line)</h5>
            <div class="chart-container">
              <canvas id="lineChart"></canvas> </div>
          </div>
        </div>
      </div>
      <script>
        document.querySelector('.toggle-btn').addEventListener('click', function () {
          document.getElementById('sidebar').classList.toggle('collapsed');
```

```
    document.getElementById('mainContent').classList.toggle('expanded');
});
const companyCount   = {{ company_count|default:0 }};
const usersCount     = {{ users_count|default:0 }};
const complaintCount = {{ complaint_count|default:0 }};
const replyCount     = {{ reply_count|default:0 }};
const resolvedCount  = Math.max(0, complaintCount - replyCount);
const chartDefaults = {
    responsive: true,
    maintainAspectRatio: false,
    plugins: {
        legend: {
            labels: {
                font: { family: "'Manrope', sans-serif", size: 12 },
                color: '#1A1A1A'
            }
        }
    }
};
new Chart(document.getElementById('barChart'), {
    type: 'bar',
    data: {
        labels: ['Companies', 'Users', 'Complaints', 'Pending Replies'],
        datasets: [{
            label: 'Count',
            data: [companyCount, usersCount, complaintCount, replyCount],
            backgroundColor: [
                'rgba(242, 217, 59, 0.85)',
                'rgba(245, 226, 122, 0.85)',
                'rgba(237, 231, 201, 0.95)',
                'rgba(26, 26, 26, 0.75)'
            ],
            borderColor: ['#F2D93B', '#F5E27A', '#EDE7C9', '#1A1A1A'],
            borderWidth: 1,
            borderRadius: 8   }]
    },
    options: {
        ...chartDefaults,
        scales: {
            y: {
                beginAtZero: true,
                ticks: { color: '#666', font: { family: "'Manrope'" } },
                grid: { color: 'rgba(237,231,201,0.5)' }
            },
            x: {
                ticks: { color: '#1A1A1A', font: { family: "'Manrope'" } },
                grid: { display: false }
            }
        },
        plugins: {
            ...chartDefaults.plugins,
```

CHAPTER 11

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